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DEPARTMENT OF DEFENSE HANDBOOK

CAPACITORS,

SELECTION AND USE OF

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides selected standard capacitors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

   a. The application information and performance characteristics contained in this handbook are offered for guidance and are not to be considered as mandatory. Additional application information will be added when coordinated with the Department of Defense.

   b. Additional capacitor types will be added to this handbook as they are developed and coordinated with the Department of Defense.

3. Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, OH 43218-3990 or e-mailed to capacitorfilter@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1. SCOPE

1.1 Scope. This handbook is for guidance only and cannot be cited as a requirement. If it is, the contractor does not have to comply. This handbook consists of the following:

a. Selected standard capacitor types, for use in the design and manufacturing of Department of Defense equipment under the jurisdiction of the Department of Defense.

b. Guides for the choice and application of capacitors for use in Department of Defense equipment.

Requirements for capacitors listed in this handbook are covered in the applicable specification (see 2.1). When it has been determined that circuit requirements cannot be met by using capacitor styles or characteristics listed in the applicable specifications, the design engineer should, with the approval of the cognizant activity, select from the applicable capacitor specification styles or characteristics not listed herein.

1.2 Purpose of handbook.

a. To provide the equipment designer with a selection of standard capacitors for use in most Department of Defense applications.

b. To control and minimize the variety of capacitors used in Department of Defense equipment in order to facilitate logistic support of equipment in the field.

c. To outline criteria pertaining to the use, choice, and application of capacitors in Department of Defense equipment.

1.3 DLA Land and Maritime SpecFinder. DLA Land and Maritime Specfinder is an online tool that can be used in conjunction with this handbook. The SpecFinder is a search tool to help you find standardization documents by selecting device characteristics. If you require certain performance features but don't know which documents might meet your needs, the SpecFinder will help you narrow your search down to a few candidate documents. It may be found at: https://landandmaritimeapps.dla.mil/Programs/SpecFind/

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-20	-	Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating), Established and Non-Established Reliability, General Specification for
MIL-PRF-81	-	Capacitor, Variable, Ceramic Dielectric, General Specification for
MIL-PRF-123	-	Capacitor, Fixed, Ceramic Dielectric (Temperature Stable and General Purpose), High Reliability, General Specification for
MIL-PRF-11693	-	Capacitor, Feed Through, Radio-Interference Reduction, DC (Hermetically Sealed in Metal Cases), Established and Non-Established Reliability General Specification for
MIL-PRF-14409	-	Capacitor, Variable (Piston Type, Tubular Trimmer), General Specification for

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MIL-PRF-19978	-	Capacitor, Fixed, Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic, or Glass Cases), Established and Non-Established Reliability, General Specification for
MIL-PRF-32535	-	Capacitor, Chip, Fixed, Ceramic Dielectric (Temperature Stable and General Purpose), Extended Range, High Reliability and Standard Reliability, General Specification for
MIL-PRF-32700	-	Capacitor, Fixed, Electrolytic (Conductive Polymer Tantalum), Surface Mount, Non-Established Reliability and Established Reliability, General Specification for
MIL-PRF-39001	-	Capacitor, Fixed, Mica Dielectric, Established Reliability, General Specification for
MIL-PRF-39003	-	Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability, General Specification for
MIL-PRF-39006	-	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, Established Reliability, General Specification for
MIL-PRF-39014	-	Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability and Non-Established Reliability, General Specification for
MIL-PRF-39018	-	Capacitor, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Non-Established Reliability, General Specification for
MIL-PRF-39022	-	Capacitor, Fixed, Metallized, Paper-Plastic Film or Plastic Film Dielectric, Direct and Alternating Current, (Hermetically Sealed in Metal or Ceramic Cases), Established Reliability, General Specification for
MIL-PRF-49137	-	Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Molded, Conformal Coated and Metal Cased with Plastic End-Fill, Non-Hermetically Sealed, Established Reliability, General Specification for
MIL-PRF-49467	-	Capacitor, Fixed, Ceramic, Multilayer, High Voltage (General Purpose), General Specification for
MIL-PRF-49470	-	Capacitor, Fixed, Ceramic Dielectric, Switch Mode Power Supply (General Purpose and Temperature Stable), Standard Reliability and High Reliability, General Specification for
MIL-PRF-55365	-	Capacitor, Fixed, Electrolytic (Tantalum), Chip, Established Reliability, Non-Established Reliability, and High Reliability, General Specification for
MIL-PRF-55514	-	Capacitor, Fixed, Plastic (or Metallized Plastic) Dielectric, DC or DC-AC, in Nonmetal Cases, Non-Established and Established Reliability General Specification for
MIL-PRF-55681	-	Capacitor, Chip, Multiple Layer, Fixed, Ceramic Dielectric, Established Reliability and Non-Established Reliability, General Specification for
MIL-PRF-83421	-	Capacitor, Fixed, Metallized Plastic Film Dielectric, (DC, AC, or DC and AC), Hermetically Sealed in Metal or Ceramic Cases, Established Reliability, General Specification for

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-1131	-	Storage Shelf Life and Reforming Procedures for Aluminum Electrolytic Fixed Capacitors

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

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3. DEFINITIONS

3.1 Rating and design application terms. A list of common terms used in rating and design application of capacitors is as follows:

a. Ambient temperature. Average or mean temperature of the medium (air, gas, liquid, etc.) surrounding an object.

b. Anode. Positive electrode of a capacitor.

c. Capacitance. Property of a capacitor which determines its ability to store electrical energy in "T" time when a given voltage is applied, measured in farads, microfarads, or picofarads.

d. Capacitance tolerance. The part manufacturer's guaranteed maximum deviation (expressed in percent) from the specified nominal value at standard (or stated) environmental conditions.

e. Failure rate level (FRL). Denotes the maximum percentage of failures (per 1,000 unit hours) based on a specified level.

f. Capacitive reactance. Opposition offered to the flow of an alternating or pulsating current by capacitance, measured in ohms.

g. Capacitor. Electronic component part consisting essentially of two conducting surfaces separated by an insulating (dielectric) material. A capacitor stores electrical energy, blocks the flow of direct current, and permits the flow of alternating or pulsating current to a degree dependent on the capacitance dielectric and the frequency.

(1) Capacitor, liquid-filled. A capacitor in which a liquid impregnant occupies substantially all of the case volume not required by the capacitor element and its connections. (Space may be allowed for the expansion of the liquid under temperature variations.)

(2) Capacitor, liquid-impregnated. A capacitor in which a liquid impregnant is dominantly contained within the foil-winding and paper-winding, but does not occupy substantially all of the case volume.

(3) Capacitor, temperature-compensating. A capacitor whose capacitance varies with temperature in a known and predictable manner.

h. Cathode. Negative electrode of a capacitor.

i. DC leakage (DCL). Stray direct current of relatively small value which flows through or across the surface of solid or liquid insulation when a voltage is impressed across the insulation.

j. Dielectric. The insulating material (e.g., air, paper, mica, oil, etc.) between the plates of a capacitor.

k. Dielectric absorption. Property of an imperfect dielectric whereby all electric charges within the body of the material caused by an electric field are not returned to the field.

l. Dielectric constant. Property of a dielectric material that determines how much electrostatic energy can be stored per unit volume when unit voltage is applied. (It is the ratio of the capacitance of a capacitor filled with a given dielectric to that of the same capacitor having a vacuum dielectric.)

m. Dielectric strength. Maximum voltage that a dielectric material can withstand without rupturing. (The value obtained for the dielectric strength will depend on the thickness and type of material and on the method and conditions of test.)

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n. Dissipation factor (DF). The ratio of resistance to reactance, measured in percent.

o. Electrolyte. Current-conducting solution (liquid or solid) between two electrodes or plates of a capacitor at least one of which is covered by a dielectric film.

p. Equivalent series resistance (ESR). The square root of the difference between the impedance squared and the reactance squared.

q. Flashpoint of impregnant. The temperature to which the impregnant (liquid or solid) must be heated in order to give off sufficient vapor to form a flammable mixture.

r. Impedance (Z). Total opposition offered to the flow of an alternating or pulsating current, measured in ohms. (Impedance is the vector sum of the resistance and the capacitive reactance, i.e., the complex ratio of voltage to current.)

s. Impregnant. A substance, usually liquid, used to saturate paper dielectric and to replace the air between its fibers. (Impregnation increases the dielectric strength and the dielectric constant of the assembled capacitor.)

t. Inactive for new design. May be used on systems designed before, but should not be used on systems designed after the date of inactivation. Applicable to Department of Defense specifications, specification sheets, and parts covered therein.

u. Insulation resistance (IR). Direct current resistance between two conductors that are separated by an insulating material.

NOTE: Capacitors are commonly subjected to two insulation resistance tests. One test determines the insulation resistance from terminal to terminal while the other test determines the insulation resistance from one or more terminals to the exterior case or insulating sleeve.

v. Partially inactive for new design. Containing both active and inactive Department of Defense specification sheets (for Department of Defense specifications), or both active and inactive parts (for Department of Defense specification sheets).

w. Power factor (PF). The ratio of resistance to impedance, measured in percent.

x. Quality factor (Q). The ratio of capacitive reactance to resistance.

y. Radio interference. Undesired conducted or radiated electrical disturbances, including transients, which may interfere with the operation of electrical or electronic communications equipment or other electronic equipment.

z. Ripple voltage (or current). The ac component of a uni-directional voltage or current (the ac component is small in comparison with the dc component).

aa. Stability. The ability of a part to resist changes of characteristic values and coefficients.

ab. Surge voltage (or current). Transient variation in the voltage or current at a point in the circuit; a voltage or current of large magnitude and short duration caused by a discontinuity in the circuit.

ac. Temperature coefficient (TC). Change in capacitance of a capacitor per degree change in temperature. It may be positive, negative, or zero and is usually expressed in parts per million per degree Celsius (ppm/°C).

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4. GENERAL REQUIREMENTS

4.1 Choice of capacitor types. The variety of capacitor types used in any particular equipment should be the minimum necessary to obtain satisfactory performance. Where more than one type of capacitor may be used in a given application (i.e., molded mica or glass types), consideration should be given to cost and availability (use of strategic materials, multiple sources, etc.). The capacitors identified in this handbook meet all the criteria for standard types (see 1.1, 4.4, and table A-I).

4.1.1 Reliability. Where quantitative reliability requirements specified as part of the equipment requirements are such that the use of parts with established reliability (ER) is dictated, such parts should be selected from the ER specification.

4.1.2 Qualified sources. After a preliminary selection of the desired capacitor has been made, reference should be made to the applicable qualified products list for listing of qualified sources.

4.2 Item identification. A type designation for any capacitor referenced herein may be constructed as indicated in the example given in the applicable section. The Part Identification Number (PIN) designations are depicted in the applicable specification.

4.3 Conflict of requirements. This handbook provides selected standard capacitors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook cannot be cited as a requirement.

4.4 Criteria for inclusion in this handbook. The criteria for the inclusion of capacitor types in this handbook are as follows:

a. The capacitor should be the best type available for general use in military equipment.

b. Coordinated Department of Defense specifications should be available (see 2.1).

c. Capacitors should be in production, or should have been in production.

5. DETAILED REQUIREMENTS

5.1 Detailed requirements. The detailed requirements for standard capacitor types are contained in the applicable specification listed in this handbook.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. General application notes are as indicated in the appendix.

6.2 Subject term (key word) listing.

Anode
Capacitive reactance
Cathode
Dielectric absorption
Power factor
Quality factor

6.3 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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APPENDIX A

GENERAL APPLICATION INFORMATION

A.1 SCOPE

A.1.1 Scope. The application information in this handbook is designed to help in the selection of specified capacitors (application information pertaining to specific capacitor types is contained in the applicable sections). As with other types of components, the most important thing a user must decide is which of the numerous types of capacitors will be best for use in the military equipment being designed. Proper selection in its broadest sense is the first step in building reliable equipment. To properly select the capacitors to be used, the user must know as much as possible about the types from which to choose. The advantages and disadvantages should be known, as well as their behavior under various environmental conditions, their construction, and their effect on circuits and the effect of circuits on them, and knowledge of what makes capacitors fail. The information contained herein is intended for guidance only.

A.1.1.1 Capacitor usage. Capacitors are used as energy-storage components to accumulate energy through long periods of time and to discharge the energy over longer or shorter periods. Parallel RC circuits will maintain bias in a circuit for long periods and, as in filter circuits, will smooth out pulsating direct current. By-pass capacitors are used to prevent the flow of direct current without impeding the flow of alternating current and attenuate low frequency currents while permitting higher frequency currents to pass. In combination with resistors, capacitors are used to reduce radio interference caused by arcing contacts, and to increase the operational life of the contacts.

A.1.1.2 Capacitor types. All capacitors, of the types widely used in electronic equipment, can be grouped into one of six basic types. They are glass and mica, electrolytic, paper and plastic, ceramic, air, and vacuum. These basic types differ from each other in size, cost, capacitance, and general characteristics. Some are better than others for a particular purpose; no one type has all of the best characteristics. The choice among them, therefore, depends on the requirements, both initial and long-term, the environment in which they must exist, and numerous other factors which the designer must understand. The designer must realize that the summaries of the requirements of a particular application must be taken into consideration and compared with the advantages and drawbacks of each of the several types, before a final choice is made. Table A-I shows the Department of Defense capacitor specification types covered in this handbook while table A-II provides a detailed selection guide for the capacitors.

A.1.2 Environmental effects on characteristics and life. The characteristics and life of all capacitors are dependent on the environments to which capacitors are exposed. Effects of various environmental conditions on capacitors are as follows:

A.1.2.1 Temperature:

a. The temperature at which the dielectric operates is a function of the ambient temperature in which the capacitor is located; the heat which is radiated or conducted to the capacitor; the internal heating of the capacitor due to I^2R losses in the conductors and dielectric; the physical construction and thermal conductivity of the materials inside the capacitors; the transfer of heat internally by conduction and convection to the container; and the heat lost from the container by convection, conduction, and radiation.

b. The insulation resistance decreases as the temperature increases. The power factor is a complex function of temperature. With polarized dielectrics, temperature-frequency combinations exist where there are large increases in power factor. This may not present any difficulties at low temperatures, since internal heating will raise the dielectric temperature and lower the power factor. An increase in power factor at high temperatures may cause thermal instability and must be considered.

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c. The capacitance of polarized dielectrics is a complex function of temperature, voltage, and frequency; nonpolarized dielectrics exhibit less change than polarized materials. It is to be noted that as the ambient temperature is decreased, many dielectrics will exhibit a very large decrease in capacitance with a relatively small change in temperature. The increased power factor at this temperature may raise the dielectric temperature sufficiently to recover the lost capacitance; however, it must be considered that when the capacitor is initially energized while at low temperatures, the capacitance will be a small percentage of its nominal value, and if the internal heating is effective, the thermal time constant of the capacitor must be considered. A change in the distance between the conductors and the effective area of the conductor due to dimensional changes will cause a change in capacitance.

d. Dielectric strength for each dielectric material is related to dielectric material and test temperature.

e. The life of a capacitor, in general, decreases with an increase in temperature. Life as a function of operating temperature is a complex function and should be determined from life-test data. In the absence of this data, the familiar 10°C rule for a chemical reaction may be used as a rough approximation. This rule states that the life decreases by a factor of two for each 10°C rise in temperature. This rule, however, should never be used outside of the temperature range specified by the manufacturer, since chemical reactions of an entirely different nature may take place at extreme temperatures. This rule should not be applied to liquid and gaseous dielectric without further investigation.

f. The operating temperature and changes in temperature also affect the mechanical structure in which the dielectric is housed. The terminal seals utilizing elastic materials or gaskets may leak due to the set temperature characteristics. The expansion and contraction of materials with different thermal coefficients may cause leaks at joints. Electrolysis effects in glass terminals increase as the temperature increases. The increase in internal pressure of liquids and gases may cause leaks. A decrease in internal pressure due to the lowering of the temperature may cause internal arc-over.

g. If the capacitor is operated in the vicinity of a component operating at high temperature, the flashpoint of the impregnant should be considered.

A.1.2.2 Pressure:

a. The dielectric strength of gases is a function of pressure, temperature, frequency, and humidity. Hermetically-sealed units must have terminals designed to operate satisfactorily at the required pressure.

b. The heat loss by convection of a capacitor is a function of pressure and must be considered.

c. Reduced pressure may produce leaks in hermetically-sealed units. An increase in pressure on the container of rolled capacitors in rectangular containers may increase the capacitance by decreasing the distance between the conductors.

A.1.2.3 Shock and vibration. The capacitors and mounting brackets, when applicable, must be of a design which will withstand the shock and vibration requirements of the particular application.

A.1.2.4 Moisture. Moisture in the dielectric will decrease the dielectric strength, life, and insulation resistance, and increase the power factor of the capacitor. In general, capacitors which operate in high humidity should be hermetically sealed. The effect of moisture on pressure contacts which are not gas-tight may result in a high resistance or open contact.

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A.1.2.5 Aging. Capacitor aging is a term used to describe the negative, logarithmic capacitance change that takes place in ceramic capacitors with time. As one might expect, the more stable dielectrics have the lowest aging rates.

Temperature compensating dielectrics, such as MIL-PRF-20 and MIL-PRF-55681 components with a characteristic of 0 ± 30 ppm/°C, over the operating temperature range of -55°C to +125°C, do not appear to age at all; however, all ceramic capacitors with high dielectric constants display an aging characteristic.

General purpose dielectrics, particularly those with a capacitance change of ± 15 percent (or greater) over the -55°C to +125°C operating temperature range, comprise this high dielectric constant family and represent the group we are concerned with.

High K ceramic dielectrics with a barium-titanate formula exhibit a phenomenon known as Curie Point crystal-phase transformation. Simply stated, most of the tiny crystals that make up the ceramic micro-structure are of cubic symmetry at a temperature of +120°C and above. Below +120°C, these same crystals take on a tetragonal shape. The specific relationship between this crystal-phase transformation and aging is not clearly understood, but it is known that they are directly related. As the crystals change from cubic to tetragonal shape, stresses are set up in the dielectric and are subsequently relieved gradually. This electrical "aging" phenomenon seems to follow the same logarithmic patterns observed in mechanical models of stress relief. Each time the capacitor is heated to approximately +120°C (Curie Point), all of the negative capacitance change that may have taken place is recovered. Upon cooling, the aging cycle begins again. This recovery process is commonly referred to as "de-aging". The entire process of aging and de-aging is predictable and can be repeated infinitely.

Another important parameter that affects capacitor aging is the application of polarizing voltage. The application of a dc voltage approximately equal to the capacitor's rating will cause an abrupt negative capacitance change; however, when the voltage is removed, the capacitor does not return to its original polarized value. If this exercise was performed on a capacitor with a known aging characteristic and the results were plotted, the resultant curves would resemble those on figure A-1.

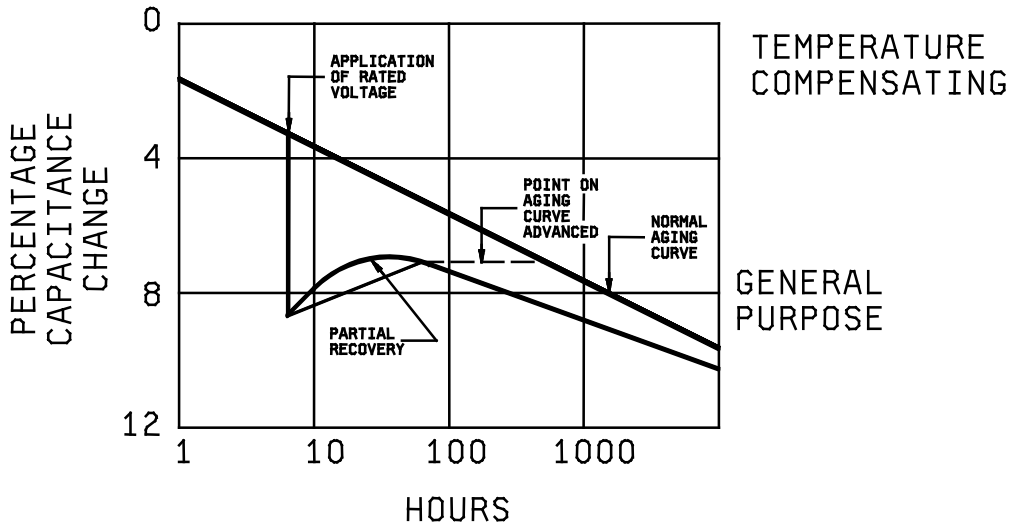


FIGURE A-1. Capacitor aging curves.

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The dc voltages and subsequent dielectric polarization of the capacitor micro-structure serve to relieve some of the stresses in the dielectric. This moves the point on the aging curve forward approximately 1.5 decades.

Most general purpose state-of-the-art dielectrics found in industry have aging rates varying from 1.5 percent to 4 percent.

In summary, the following points should be kept in mind when dealing with the phenomenon of ceramic capacitor aging:

a. The process is completely repeatable and predictable.

b. Capacitance change is negative and logarithmic with respect to time.

c. Application of dc bias can move a point on the curve forward in time.

This wide capacitance change, as a result of "shelf" aging and temperature cycling, illustrates why tight-tolerance high K ceramics are not common in the electronics industry.

A.1.2.6 Capacitor tests. The average component is a complex device. For this reason it is impossible for the manufacturer to guarantee an exact minimum life of an individual capacitor; for example, under any given circuit or environmental condition of usage, all he can do is to provide statistical guides as to the probable minimum life or reliability of the unit when considered as a member of a large family of units.

A.1.2.7 Capacitor misuse. A capacitor may fail when subjected to environmental or operational conditions for which the capacitor was not designed or manufactured. The designer must have a clear picture of the safety factors built into the units, of the safety factors he adds of his own accord, and of the numerous effects of circuit and environmental conditions on the parameters. It is not enough to know only the capacitance and the voltage rating. It is important to know to what extent the capacitance varies with environment; how much the internal resistance of the capacitor varies with temperature, current, voltage, or frequency; of the effects of all of these factors on insulation resistance, breakdown voltage, and other basic capacitor characteristics which are not essential to the circuit but which do invariably accompany the necessary capacitance.

A.1.3 Principal applications. Some of the principal applications of the various types of capacitors are shown in table A-II.

A.1.4 Capacitor selection. The designer, in selecting a capacitor type for a particular function to be performed, must weigh numerous factors before coming to a final decision. Selection normally starts with the most important characteristic for the application, then selecting and compromising other characteristics.

A.1.4.1 Selection factors. The most important of these factors are noted below with some of the reasons why these factors are important.

A.1.4.1.1 Temperature effects:

a. Capacitance:

(1) By variations in dielectric constant.

(2) By changing conductor area or spacing.

b. Leakage current, through change in specific resistance.

c. Breakdown voltage at high temperatures and effect of frequency on heating.

d. Current rating, when affected by heating.

e. Oil, gas, or electrolyte leakage through seals.

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A.1.4.1.2 Humidity effects:

a. Leakage current.

b. Breakdown voltage.

c. Effect on power factor or Q.

A.1.4.1.3 Barometric pressure effects:

a. Breakdown voltage.

b. Oil, gas, or electrolyte leakage through seals.

A.1.4.1.4 Applied voltage effects:

a. Leakage current.

b. Heating and its accompanying effects.

c. Breakdown of dielectric; effect of frequency.

d. Corona.

e. Insulation to case or chassis.

A.1.4.1.5 Vibration effects:

a. Capacitance change through mechanical vibration.

b. Mechanical distortion of elements, terminals, or case.

A.1.4.1.6 Current effects:

a. Effect on internal temperature rise and life of capacitor.

b. Ability of conductors to carry currents from a thermal viewpoint.

A.1.4.1.7 Life. Affected by all environmental and circuit conditions.

A.1.4.1.8 Stability. Also affected by all environmental and circuit conditions.

A.1.4.1.9 Retrace. After a capacitance change.

A.1.4.1.10 Size, volume, cost, and mounting method.

A.1.4.2 Capacitor selection chart. Table A-I and table A-II list the capacitor styles available in each specification represented in this standard. The data given is approximate and is meant as an aid in selecting capacitors only.

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TABLE A-I. Capacitor categories.

CERAMIC	MIL-PRF-20	Temperature Compensating (ER and Non-ER)
	MIL-PRF-81	Variable (Non-ER)
	MIL-PRF-123	Temperature Stable and General Purpose (High Reliability)
	MIL-PRF-32535	Chip, Temperature Stable and General Purpose (High Reliability and Standard Reliability)
	MIL-PRF-39014	General Purpose (ER and Non-ER)
	MIL-PRF-49467	Multilayer High Voltage
	MIL-PRF-49470	Switch Mode Power Supply (High Reliability)
	MIL-PRF-55681	Chip (ER)
ELECTROLYTIC	MIL-PRF-32700	Conductive Polymer Tantalum Chip (ER and Non-ER)
	MIL-PRF-39003	Solid Electrolyte Tantalum (ER)
	MIL-PRF-39006	Nonsolid Electrolyte Tantalum (ER)
	MIL-PRF-39018	Aluminum Oxide (ER)
	MIL-PRF-49137	Solid Electrolyte Tantalum (Non-ER)
	MIL-PRF-55365	Tantalum Chip (ER and Non-ER)
GLASS	MIL-PRF-14409	Variable Piston Type Tubular Trimmer (Non-ER)
MICA	MIL-PRF-39001	(ER)
PAPER / PLASTIC	MIL-PRF-11693	Feed Through, Radio-Interference Reduction, DC (ER and Non-ER)
	MIL-PRF-19978	(ER)
	MIL-PRF-39022	DC/AC (ER)
	MIL-PRF-55514	DC or DC-AC (ER)
	MIL-PRF-83421	DC/AC, Hermetically Sealed (ER)

A.1.5 Application data. The following should be considered in the selection and use of a capacitor type:

a. The capacitance tolerance that the circuit designer uses in order to design a circuit which will operate satisfactorily for the desired time requires (1) acceptable tolerances according to specification; (2) capacitance-temperature characteristics; (3) capacitance-voltage characteristics; (4) retrace characteristics; (5) capacitance-frequency characteristics; (6) dielectric absorption; (7) capacitance as a function of pressure, vibration, and shock; and (8) capacitor aging in the circuit and shelf storage.

b. Capacitance between the capacitor terminals and case may be a consideration, as will stray capacitance and leakage currents. The terminal connected to the outside conductor is often identified by the manufacturer so that the circuit can minimize these effects.

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c. The capacitance-temperature characteristic can be compensated for by using more than one type of capacitor to obtain the required capacitance. The characteristics of other circuit components may also be used for compensation.

d. The peak voltage which is applied to the capacitor should not exceed the rating in the applicable specification. The safety factor between the peak applied voltage, the test voltage, and the breakdown voltage is of a statistical nature. The same peak voltage, in general, may decrease with (1) aging, (2) an increase in temperature, (3) an increase of area of dielectric, (4) higher frequencies of applied voltage, (5) a decrease in pressure, or (6) the entrance of moisture into the capacitor. In many applications, it is necessary to derate the capacitor from the specified voltage to provide the desired performance for the required time. It is to be emphasized that short-duration transient voltages cannot be neglected in capacitor applications.

e. The use of the self-healing properties of certain types of capacitors may not be desirable in circuits where intermittent failures and noise would be troublesome. Some types are not self-healing at low voltages.

f. Operation of capacitors above the corona-starting voltage will reduce the life and will produce noise. Liquid-impregnated dielectrics have a higher corona-starting voltage than dry solid dielectrics.

g. When a capacitor is operated at high voltages above ground, and when it is insulated from ground with supplementary insulation, one terminal should be connected to the case, since the division of voltage depends on capacitance between capacitor rolls and case and the capacitance between case and chassis.

h. The peak charge and discharge currents must be considered on the basis of the time constant of the circuit.

i. Internal heating and ambient temperature must be considered.

j. To determine the surface temperature rise of a capacitor, multiply the volt-amperes supplied to the unit by the power factor. This gives the watts lost in the capacitor. Dividing the watts lost by the surface area in square inches and referring to figure A-2 will give the approximate surface temperature rise.

k. Environmental conditions such as humidity, pressure, corrosive atmospheres, fungus growth, shock, and vibration must be considered.

l. The insulation resistance must be considered, especially at high temperatures.

m. In series operation on dc, balancing resistors should be considered.

n. The effective inductance of a large capacitor can be reduced by shunting it with a small capacitor.

o. The inductance of various types of capacitors varies over wide limits.

p. Since capacitors have inductance, the operation of capacitors in parallel in circuits with fast rise times or transients may result in transient oscillations.

q. Poor electrical contacts may open at low voltages and be noisy.

r. The stored energy in capacitors can be dangerous to personnel and equipment and suitable precautions should be taken.

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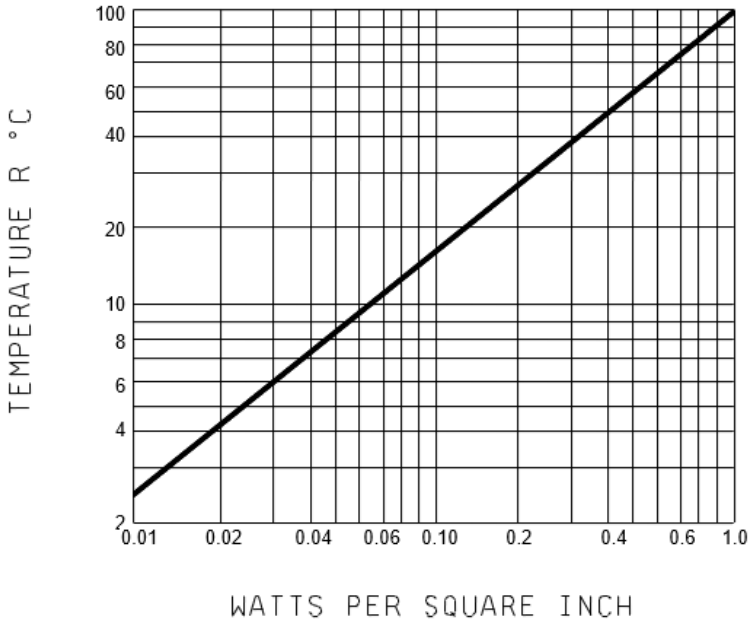


FIGURE A-2. Temperature rise as a function of watts per square-inch-power dissipation.

s. Oil-filled or acid-filled units should not be subjected to severe mechanical stresses. Leakage of the fluid can destroy the capacitor together with adjacent components.

t. Liquid-filled units should not be used inverted because internal corona may result.

u. Nonhermetically-sealed capacitors may be pervious to moisture by the process of "breathing."

v. Capacitors for ac and pulse operation require special ratings and tests; these are not covered in most military specifications.

A.2. APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-83421/1	-	Capacitor, Fixed, Metallized Plastic Film, Dielectric, DC and AC, Hermetically Sealed in Metal Cases, Established Reliability, Styles CRH01, CRH02, CRH03, CRH04, CRH05, CRH06, CRH07, CRH08, CRH09, and CRH00
MIL-PRF-83421/6	-	Capacitor, Fixed, Metallized Plastic Film Dielectric, DC and AC, Hermetically Sealed in Metal Cases, Established Reliability, Styles CRH31, CRH32, CRH33, CRH34, and CRH35

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-690	-	Failure Rate Sampling Plans and Procedures

(Copies of these documents are available online at https://assist.dla.mil/.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

IPC INTERNATIONAL (IPC)

J-STD-020	-	Moisture/Reflow Sensitivity Classification for Nonhermetic Surface Mount Devices

(Copies of these documents are available online at https://www.ipc.org/).

JEDEC – SOLID STATE TECHNOLOGY ASSOCIATION (JEDEC)

J-STD-033	-	Handling, Packing, Shipping and Use of Moisture, Reflow, and Process Sensitive Devices

(Copies of these documents are available online at https://www.jedec.org).

A.3. GENERAL CHARACTERISTICS OF CAPACITORS

A.3.1 General characteristics of ceramic capacitors.

A.3.1.1 MIL-PRF-20, CCR and CC, Capacitor, Fixed, Ceramic Dielectric (Temperature Compensating), Established Reliability and Non-Established Reliability.

A.3.1.1.1 Use. These ceramic, temperature compensating, fixed capacitors are primarily designed for use where compensation is needed to counteract reactive changes, caused by temperature variations, in other circuit components. Ceramic capacitors are substantially smaller than paper or mica units of the same capacitance and voltage rating. They can be used where mica or paper capacitors have too wide a capacitance tolerance. The lead placement makes ceramic capacitors suitable for printed-circuit use.

By using these units, frequency drift in radio frequency, oscillator, and intermediate frequency (IF) circuits due to temperature effects can be compensated individually in each circuit. In IF stages where the frequency variation is uniform, satisfactory operation can be obtained by designing the temperature-compensating capacitor into the oscillator circuit. RF circuit reactive changes caused by temperature variations cannot be compensated for in the oscillator circuit; in these cases, and where more critical tuning accuracy is required, it is necessary that compensating capacitors be inserted directly into each circuit.

In RF circuits tuned by a variable capacitor, a shunt compensating capacitor of low value and high compensating characteristics may be used. In slug-tuned circuits, the total capacitance required can be provided by using a compensating capacitor with the desired temperature coefficient. In oscillator circuits, more linear tuning can be obtained by using proper temperature coefficients in both the series and the shunt capacitances of the tank circuit.

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High insulation resistance makes these capacitors well suited to coupling applications between plate and grid circuits of electron tubes. Extremely low leakage and small physical size make them suitable for transistor circuit design. They are also useful in filter and by-pass circuits.

If possible, the temperature-time curve of the selected capacitor should be the exact opposite of the temperature-time curve of the coil (or other component) being stabilized. Combinations of different capacitance values and temperature coefficients can give more precise compensation than can be obtained from a single capacitor. Full consideration should be given to the physical placement of compensating, and compensated for, components. Locations near hot transistors will cause much greater reactive variations than spots adjacent to a cool, external chassis.

Ceramic dielectrics are frequency sensitive, both the capacitance and the capacitance change with temperature will be different at different measuring frequencies. For extremely accurate compensation, the units should be measured at the proposed operating frequency.

Capacitors meeting the established reliability (ER) requirements specified herein have a failure rate level (FRL) ranging from 1.0 percent per 1,000 hours to 0.00l percent per 1,000 hours. These FRLs are established at a 90-percent confidence level based on the life test parameters specified and are maintained at a 10-percent producer's risk. An acceleration factor of 8:1 has been used to relate the life test data at 200 percent of rated voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature.

A.3.1.1.2 Construction. Physically, the most common types of temperature-compensating, ceramic-dielectric capacitors are small monolithic tubular and rectangular types covered by insulating resin, plastic, or ceramic. Because the constituent materials have molecular polar moments, the dielectric constants of some mixes reach hundreds (even thousands) of times the value of paper, mica, and plastic films. This results in ceramic-dielectric capacitors having the largest capacitance-to-size ratios of all high-resistance dielectrics.

A.3.1.1.3 Dielectric strength. It is recommended that supplementary insulation be used where the capacitor body will normally contact parts with a potential difference of more than 750 volts.

A.3.1.2 MIL-PRF-81, CV, Capacitor, Variable, Ceramic Dielectric.

A.3.1.2.1 Use. These capacitors are small-sized trimmer capacitors designed for use where fine tuning adjustments are periodically required during the life of the equipment. Normally they are used for trimming and coupling in such circuits as intermediate frequency, radio-frequency, oscillator, phase shifter, and discriminator stages. Because of their low mass, these units are relatively stable against shock and vibration which tend to cause changes in capacitance. Where a higher order of stability is required, air trimmers should be used. Capacitance and adjustment are relatively linear.

A.3.1.2.2 Construction. Each unit consists of a single stator and a single rotor for each section, made of ceramic material impregnated with transformer or silicone oil. Pure silver is fired and burnished on the top of the base of the stator in a half-moon pattern. The rotor, usually of titanium dioxide, has pure silver contact points. The contact surfaces of both the stator and the rotor are ground and lapped flat, thus eliminating air space variations with temperature.

The principle of operation is similar to that of an air-dielectric tuning capacitor where the overlap of the stator and rotor determines the capacitance; in these units, the ceramic dielectric replaces the air dielectric. Rotors may be rotated continuously; full capacitance change occurs during each rotation. The approximate maximum capacitance point is indicated on the capacitor.

A.3.1.2.3 Mounting. These capacitors may be mounted close to a metal panel with little increase in capacitance. To avoid cracking or chipping of the ceramic mounting base, a resilient mounting (or mounting surface spacer) should be used.

A.3.1.2.4 Capacitance change with temperature. When measurements are made after the capacitors have reached thermal stability at each temperature setting (at a frequency between 0.1 and 1.2 megahertz (MHz), and with

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the capacitor set at 80 to 90 percent of maximum capacity, the changes of nominal capacitance from the value measured at +25°C may vary from -4.5 to +14.0 percent at -55°C or -10.0 to +2.0 percent at +85°C. Since the temperature sensitivity is nonlinear over the capacitance range and varies greatly between units, these capacitors should not be designed into circuits as temperature compensating units. The capacitance drift remains within 0.5 pF.

A.3.1.3 MIL-PRF-123, CKS, Capacitor, Fixed, Ceramic Dielectric, (Temperature Stable and General Purpose), High Reliability.

A.3.1.3.1 Use. These high reliability, general purpose (BX and BR) and temperature stable (BP and BG) ceramic leaded and nonleaded capacitors are intended for space, missile and other high reliability applications. Capacitors covered by this specification may be used in critical frequency determining applications, timing circuits, and other applications where absolute stability is required (BP and BG) and in applications where appreciable variations in capacitance with respect to temperature, voltage, frequency and life can be tolerated (BX and BR). Life tests in this specification are performed at 2 times rated voltage at maximum rated temperature, and an assumed acceleration factor of 8:1 is used to relate life test data obtained at 2 times rated voltage to performance at rated voltage.

A.3.1.4 MIL-PRF-32535, Capacitor, Chip, Fixed, Ceramic Dielectric (Temperature Stable and General Purpose), Extended Range, High Reliability and Standard Reliability.

A.3.1.4.1 Use. These capacitors are intended for use in standard and high reliability military applications. High reliability applications include use in space and launch vehicles. Capacitors covered by this specification are unique due to the fact that they must be able to operate satisfactorily in standard and high reliability military systems under demanding conditions. These capacitors also offer reliability that is verified under a qualification system. Commercial components are not designed to withstand these military environmental conditions.

Two product levels are offered: M level (standard reliability) and T level (high reliability). T level capacitors are intended for use in space, missile, and other high reliability applications. MIL-PRF-32535 capacitors have higher capacitance values than MIL-PRF-123 and MILPRF-55681 capacitors of the same size and voltage rating.

A.3.1.4.2 Soldering guidance.

A.3.1.4.2.1 Wave soldering. Wave soldering is not recommended for multiple layer ceramic capacitors (MLCC) larger than size 1206. Wave soldering uses liquid metal that has the highest heat transfer rate and is the hardest soldering method to use without shocking surface mount components. Extreme thermal shock is evidenced by visible cracks on the surface and sides of the capacitor. These cracks start at or near the termination and ceramic interface extending from the termination down along the capacitor edge. These surface cracks can become elliptical or circular shaped in the larger capacitor sizes. The cracks can eventually lead to failure of the capacitors. The manufacturer should be contacted for further information.

A.3.1.4.2.2 Manual soldering. Manual soldering with a soldering iron is not recommended for MLCCs. Damage due to thermal shock is common in MLCCs that are manually attached or reworked with a soldering iron. When manual soldering is absolutely necessary, hot air reflow is recommended. The manufacturer should be contacted for further information.

A.3.2.4.3 T level requirements. T level capacitors require the following in order to ensure high reliability:

a. In-process inspection consisting of a pre-termination nondestructive internal examination and a post-termination visual examination.

b. Inspection of product for delivery requires in-process, group A, and group B inspections for each production lot.

c. Enhanced production lot definition.

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d. Thermal shock and nondestructive internal examination prior to voltage conditioning in the group A inspection followed by 100 percent IR (at +125°C) screening.

e. Destructive physical analysis.

f. Tightened percent defective allowed (PDA) for qualification (group 1) and group A (subgroup 1).

g. Enhanced thermal shock and voltage conditioning testing requirements.

h. Enhanced life testing requirements.

A.3.1.5 MIL-PRF-39014, CKR, Capacitor, Fixed, Ceramic Dielectric (General Purpose), Established Reliability and Nonestablished Reliability.

A.3.1.5.1 Use. These capacitors are primarily designed for use where a small physical size with comparatively large electrical capacitance and high insulation resistance is required. Ceramic capacitors are substantially smaller than paper or mica units of the same capacitance and voltage rating. General-purpose ceramic capacitors are not intended for precision use but are suitable for use as by-pass, filter, and noncritical coupling elements in high-frequency circuits where appreciable changes in capacitance, caused by temperature variations, can be tolerated. These units are not recommended for use directly in frequency-determining circuits. Typical recommended applications include resistive-capacitance coupling for audio and radio frequency, RF and intermediate frequency cathode bypass, automatic volume control filtering, tone compensation, volume-control RF bypass, antenna coupling, and audio-plate RF bypass. All of these applications are of the type where dissipation factor is not critical, and moderate changes due to temperature, voltage, and frequency variations do not affect the proper functioning of the circuit. For example: An emitter bypass for 100 megahertz (MHz), having a nominal capacitance of 680 picofarads (pF), will give a capacitance reactance of 2.34 ohms. Since this reactance is very small compared with the emitter resistor, there would be no measurable effect on the 2.34-ohm value if the capacitance should change by several percent due to a temperature variation, nor would a dissipation of 4 percent be noticeable.

Disk and thin-plated subminiature types are extremely compact and have an inherent low-series inductance due to their construction. The placement of the leads facilitates making close-coupled low-inductance connections and these capacitors are suitable for printed-circuit applications. High insulation resistance allows these capacitors to be used in vacuum-tube grid circuits; their extremely low leakage and small physical size make them suitable for use in transistor circuitry.

During circuit design, consideration should be given to the changes in dielectric constant caused by temperature, electric field intensity, applied frequency, and shelf aging.

Established reliability (ER) capacitors covered by this specification have failure rate levels (FRL) (M, P, R, and S) ranging from 1.0 to 0.001 percent per 1,000 hours. These FRLs are established at a 90-percent confidence level and maintained at a 10-percent producer's risk and based on life tests performed at maximum rated voltage at maximum rated temperature. An acceleration factor of 8:1 has been used to relate life test data obtained at 200 percent of rated voltage at maximum rated temperature, to rated voltage at rated temperature.

A.3.1.5.2 Humid operating conditions. Ceramic dielectric materials are effectively impermeable, and have practically no moisture absorption even after considerable exposure to humid conditions. Thus, these units are intended to operate, through their full temperature range, at relative humidity up to 95 percent. Nevertheless, the termination materials under moisture conditions are subject to ionic migration that can cause capacitor failure (see A.3.1.4.8).

A.3.1.5.3 Construction. A ceramic capacitor consists of a ceramic dielectric on which a thin metallic film, usually silver, has been fired at very high temperatures. Terminal leads are attached to the electrodes by a pressure contact or by soldering. Ceramic capacitors are encapsulated to protect the dielectric from the environment and to electrically insulate the capacitor. The disk types are covered by an insulating resin, plastic, or ceramic; the thin-plated subminiature types may be dipped, molded, or placed into preformed cases. Because the constituent materials have molecular polar moments, the dielectric constants of some mixes reach hundreds (even thousands), of times the

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value of paper, mica, and plastic films. This results in ceramics having the largest capacitance-to-size ratios of all high-resistance dielectrics.

A.3.1.5.4 Soldering. Care should be used in soldering the leads. Excessive heat may damage the encapsulation and weaken the electrode to terminal lead contact. Sudden changes in temperature, such as those experienced in soldering, can crack the encapsulation or the ceramic dielectric. Leads should not be bent close to the case nor should any strain be imposed on the capacitor body to avoid fracturing the encapsulation or ceramic dielectric.

A.3.1.5.5 Dissipation factor. For the recommended applications, the dissipation factor is negligibly low. The power factor decreases as temperature is increased; this provides an advantage where operation above room temperature is required.

A.3.1.5.6 Case insulation. It is not intended that the case insulation be subjected to sustained voltage in excess of 150 percent of the dc rated voltage of the capacitor. Supplementary insulation should be provided where the case may come in contact with higher voltage.

A.3.1.5.7 Capacitance as a function of operating conditions. The dielectric constant of these capacitors exhibits a considerable dependence on field strength. Large variations in capacitance may be experienced with changes in ac or dc voltages. The dielectric constant may decrease with time and may be as low as 75 percent of the original value after 1,000 hours. The dielectric constant is dependent on frequency and decreases as the frequency is increased; it also decreases with temperature.

A.3.1.5.8 Silver migration. When silver electrodes in the ceramic capacitor are exposed to high humidities and high dc potentials, silver ion migration may take place and short circuit capacitors after relatively short periods of time. Excessive moisture during periods of storage should be avoided since the encapsulation material may absorb moisture and silver ion migration may occur when the capacitors are later put into service.

A.3.1.6 MIL-PRF-49467, HV, Capacitor, Fixed, Ceramic, Multilayer, High Voltage (General Purpose). These capacitors are primarily designed for use where a small physical size with comparatively large electrical capacitance and high insulation resistance is required. General purpose (BR and BZ characteristics) ceramic capacitors are not intended for frequency-determining or precision circuits but are suitable for use as by-pass, filter, and noncritical coupling elements in high-frequency circuits. All of these applications are of the type where dissipation factor is not critical, and moderate changes due to temperature, voltage, and frequency variations do not effect the proper functioning of the circuit. BP characteristic ceramic capacitors are for use in critical frequency determining applications, timing circuits, and other applications where absolute stability is required.

A.3.1.7 MIL-PRF-49470, PS, Capacitor, Fixed, Ceramic Dielectric, Switch Mode Power Supply (General Purpose and Temperature Stable). These ceramic capacitors are primarily designed for use in switch mode power supplies. General purpose (BQ, BR, and BX characteristics) ceramic capacitors are not intended for frequency-determining or precision circuits but are suitable for use as by-pass, filter, and noncritical coupling elements in high-frequency circuits. All of these applications are of the type where dissipation factor is not critical and moderate changes due to temperature, voltage, and frequency variations do not affect the proper functioning of the circuit. BP characteristic ceramic capacitors are for use in critical frequency determining applications, timing circuits, and other applications where absolute stability is required. An acceleration factor of 8:1 has been used to relate life test data obtained at 200 percent of rated voltage at maximum rated temperature to rated voltage at rated temperature.

A.3.1.8 MIL-PRF-55681, CDR, Capacitor, Chip, Multiple Layer, Fixed, Ceramic Dielectric, Established Reliability and Non-Established Reliability.

A.3.1.8.1 Use. These ceramic chip capacitors are intended to be used in thin and thick film hybrid circuits where micro-circuitry is indicated for filter by-pass coupling applications, and where variation in capacitance with respect to temperature, voltage, frequency, and life can be tolerated. This specification also covers another ER capacitor, using ceramic dielectric, primarily intended for use in resonant circuits with high Q factor and stability of capacitance with respect to temperature, frequency, and life.

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ER capacitors covered by this specification have FRLs ranging from 1.0 percent to 0.001 percent per 1,000 hours. These FRLs are established at a 90-percent confidence level and maintained at a 10-percent producer's risk and are based on life tests performed at maximum rated voltage at maximum rated temperature. An acceleration factor of 8:1 has been used to relate life test data obtained at 200 percent of rated voltage at maximum rated temperature, to rated voltage at rated temperature.

A.3.1.8.2 Ambient operating conditions. Designers are cautioned to give consideration to the change in dielectric constant with temperature, shelf aging, and electric-field intensity, and should recognize that the insulation resistance may vary with humidity and organic contamination of the ceramic chip surfaces.

A.3.1.8.3 Metallized terminations. It should be noted that when pure silver is used for the terminations, silver migration between the terminations may occur under conditions of simultaneous application of high humidity and dc voltage. This produces a troublesome electrical leakage path across the capacitor chip. Addition of about 20 percent of palladium to the silver to form an alloy will retard the tendency toward silver migration. Complete overcoating of the silver termination by the lead-tin bonding solder also will retard the tendency toward silver migration. Addition of about 3 percent of silver to the lead-tin bonding solder will tend to reduce the leaching of the silver from a silver termination during the solder bonding operation.

The termination finish table in MIL-PRF-55681 has as a choice base metallization-barrier under the termination finish. The base metallization-barrier is a nickel barrier.

A.3.1.8.4 Effect of mounting reliability. Voltage temperature limits, resistance to thermal shock, and reliability may be affected as a result of mounting on substrates with dissimilar coefficients of expansion from capacitor material. Care should be taken in the selection of substrate material.

A.3.1.8.5 Wave soldering of large chips. Chips larger than 1206 are not recommended for wave soldering. Wave soldering uses liquid metal that has the highest heat transfer rate and is the hardest soldering method to use without shocking surface mount components. When extreme thermal shock is present, it is very obvious with visible cracks on the surface and sides of the capacitor. These cracks start at or near the termination and ceramic interface extending from the termination down along the capacitor edge. These surface cracks can become elliptical or circular shaped in the larger capacitor sizes. The cracks can eventually lead to failure of the capacitors. Some of this information was taken from MIL-PRF-55681 manufacturers Internet sites. The manufacturers should be contacted for further information.

A.3.2 General characteristics of electrolytic capacitors.

A.3.2.1 MIL-PRF-32700, Capacitor, Fixed, Electrolytic (Conductive Polymer Tantalum), Surface Mount, Non-Established Reliability and Established Reliability.

A.3.2.1.1 Use. These tantalum chip capacitors are intended to be used in surface mount applications. These capacitors are not hermetically sealed or moisture proof. These capacitors are suited to harsh environment applications under 20 Gs of high frequency vibration and over a wide temperature range (-55°C to +125°C). These capacitors also offer reliability grading that is verified under a qualification system. Many commercial components are not designed to withstand these military environmental conditions.

The ER capacitors have calculated failure rate levels (FRLs) based on a reliability assessment performed on each production lot. FRLs are of the form percentage of failures per 1,000 hours at dc rated voltage and +85°C.

A.3.2.1.2 Soldering heat. Caution should be exercised when subjecting these sample units to soldering heat. Preheat and soldering exposure times and temperatures should be held to a minimum.

A.3.2.1.3 Moisture and handling sensitivity. These conductive polymer tantalum capacitors may be sensitive to moisture induced stress if exposed to humidity for extended periods of time. J-STD-020 (Moisture/Reflow Sensitivity Classification for Nonhermetic Surface Mount Devices) provides a procedure to identify the moisture sensitivity level rating for non-hermetic surface mount devices so that they can be properly packaged, stored, and handled. The MSL ratings for the capacitors qualified to this specification are based on the tin/lead reflow profile specified in J-STD-020.

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If humidity exposure exceeds J-STD-020 limits, users may consult J-STD-033 (Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices) for general bake out and handling procedures; however, individual capacitor manufacturers may have more specific recommendations, especially for those capacitors packaged in tape and reel. Passing the criteria specified in J-STD-020 is not sufficient by itself to provide assurance of long-term reliability.

A.3.2.1.4 Solder reflow. These capacitors may become susceptible to damage during solder reflow due to moisture absorption. Vapor pressure of moisture inside a non-hermetic package increases greatly when exposed to the high temperature of solder reflow, potentially resulting in internal cracking or damage to the capacitor. In the most severe cases, the internal pressure can cause external package cracks, commonly referred to as the “popcorn” phenomenon.

A.3.2.1.5 ESR drift. ESR will drift upward over time in an oxygen environment due to oxidation. The higher the temperature exposure, the faster the ESR will drift. Users should contact the manufacturer for drift characteristics. The ESR of capacitors qualified to this specification may also increase due to moisture absorption. Moisture can penetrate the non-hermetic package and be absorbed by the cathode elements. This absorption can reduce conductivity of the contact points which increases the resistance.

A.3.2.1.6 Anomalous charging current (ACC). Conductive polymer tantalum capacitors may be susceptible to anomalous charging currents. Anomalous charging current is a higher amount of current flowing into a capacitor while charging than would be predicted by C dv/dt + reasonable leakage current. This can occur when the capacitor is in a dry state (loss of water bound to polymer), such as after reflow soldering or in a vacuum. Users should contact the manufacturer for ACC characteristics.

A.3.2.1.7 Capacitance decrease in dry conditions. These capacitors may be susceptible to a decrease in capacitance in dry conditions. Users should contact the manufacturer for more information.

A.3.2.1.8 Derating. It is recommended that capacitors with voltage ratings up to 10V be derated by 10 percent (90 percent of rated voltage) and capacitors with voltage ratings greater than 10V be derated by 20 percent (80 percent of rated voltage). Users should contact the manufacturer for more information.

A.3.2.2 MIL-PRF-39003, CSR, Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability.

A.3.2.2.1 Use. These tantalum capacitors are intended mainly for use in filter, bypass, coupling, blocking, and other low-voltage applications (such as transistor circuits) where stability, size, weight, and shelf life are important factors. The dc leakage and dissipation factor of the suggested unit should be taken into consideration when designing transistor, timing, phase-shifting, and vacuum tube grid circuits. Operation of capacitors in parallel increases the risk of dc surge current failures in low-impedance circuits. The user is cautioned that the energy stored in a parallel capacitor circuit may discharge through other capacitors in the circuit. The life of these capacitors is primarily dependent on voltage and temperature. These capacitors should not be used above the derated voltage at maximum rated temperature, +125°C. The FRL at +125°C is not established in this specification; however, proof tests at +125°C are required.

Rated voltages range from 6 to 100 volts dc with surge voltages of 8 to 130 volts dc, respectively. These capacitors are designed for full rated voltage operation between -55°C and +85°C, derated linearly between +85°C and +125°C to the value shown in table I of the specification. These capacitors have reliability ratings established on the basis of life tests performed at specified voltage at +85°C for FRLs ranging from 1.0 percent per 1,000 hours to 0.001 percent per 1,000 hours in accordance with MIL-STD-690. These FRLs are established at a 60-percent confidence level and are maintained at a 10-percent producer's risk for Exponential distribution and 0.1 percent per 1,000 hours to 0.001 percent per 1,000 hours at a 90-percent confidence level for Weibull distribution.

These capacitors are available as polarized and nonpolarized types. Polarized types should have their cases at the same potential as the negative lead; they should be used only in dc circuits with polarity observed. Nonpolarized types should be used where reversal of potential occurs.

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A.3.2.2.2 Construction. A porous tantalum pellet or wire serves as the anode of a solid tantalum capacitor. The surfaces of the anode are electrochemically converted to an oxide of tantalum which serves as the dielectric. These surfaces are coated with an oxide semiconductor which is the working electrolyte in solid form. This oxide semiconductor establishes contact with all of the complex surfaces of the anodized pellet and is capable of healing imperfections of the tantalum oxide dielectric film.

NOTE: In high impedance circuits, momentary breakdowns (if present) will self-heal; however, in low impedance circuits, their self-healing characteristics under momentary breakdown of the dielectric film will be nonexistent. The large currents in low impedance circuits will cause permanent damage to the capacitor.

A.3.2.2.3 Voltage derating. When properly derated, these units may be operated over a temperature range of -55°C to +125°C. The derated voltage at +125°C is approximately 66 percent of the full rated voltage. While this capacitor type is designed for full rated voltage between -55°C and +85°, historical application has also included a recommended minimum de-rating of 50% rated voltage.

A.3.2.2.4 Surge current. Surge current testing is performed to verify the ability of capacitors to withstand high current peaks at rapid turn-on/off conditions in specific applications. It has been found to be very effective in screening out this potential failure mechanism. The surge current testing is optional. The specific tests and options are detailed in MIL-PRF-39003.

A.3.2.2.5 Reverse voltage. These capacitors are capable of withstanding peak voltages in the reverse direction equal to 15 percent of their dc rating at +25°C; 10 percent at +55°C; 5 percent at +85°C; and 1 percent at +125°C.

A.3.2.2.6 Permissible ripple voltage. These capacitors may be operated with an impressed ripple (ac) voltage provided the capacitors do not exceed their heat-dissipation limits. Total heat-dissipation limits depend on the ambient operating temperature and the operating frequency. For example: A 10 μF capacitor of any voltage may be operated at 1.9 V_{rms}, 120 Hz, +25°C; or at 0.75 V_{rms}, 120 Hz, +125°C. When this same capacitor is subjected to a ripple frequency of 1,000 Hz, the permissible ripple voltage must be reduced by the ratio of permissible ac at 120 Hz as follows: 1.9*(0.47/1.9) = 0.47V_{rms} at +25°C, 1,000Hz or 0.75*(0.47/1.9) = 0.19V_{rms} at +125°C, 1,000Hz. The sum of the applied dc bias voltage and the peak of the ac ripple voltage should not exceed the dc rated voltage for the applicable ambient temperature. Permissible ac voltage determined may be applied when the dc voltage is zero or near zero, provided the negative peak of the ac voltage does not exceed the allowable reverse voltage limits of 1 percent of the rated voltage at +125°C. For CSR21 capacitors, ripple voltage is more often limited by restraints on reversal of voltage. Ripple current limitations are more significant because the degradation mode is thermal and must not be allowed to exceed the maximum levels specified for each rating, frequency, and ambient temperature.

A.3.2.2.7 Series and parallel networks. It is recommended that when these capacitors are connected in series, the maximum voltage across the network should not be greater than the lowest voltage rating of any capacitor in the network, or that voltage divider resistors be used to prevent over voltage on one or more units of the series capacitor group.

To obtain a higher capacitance than can be obtained from a single capacitor, a number of units may be connected in parallel. However, the sum of the peak ripple and the applied dc voltage should not exceed the dc working voltage of the unit with the lowest voltage rating. The connecting leads of the parallel network should be large enough to carry the combined currents without reducing the effective capacitance due to series lead resistance.

A.3.2.2.8 Comparison with aluminum electrolytics. Tantalum solid electrolytic capacitors differ from aluminum electrolytics in several important aspects; namely, substantially indefinite shelf life, superior low temperature characteristics, complete freedom from electrolyte leakage, and higher operating temperatures. However, because tantalum electrolytic capacitors generally are more costly than aluminum electrolytic capacitors, consideration should be given to the use of aluminum electrolytic capacitors if their performance characteristics and physical sizes are suitable and if the application will permit.

A.3.2.2.9 Mounting. Supplementary mounting means should be used where the application of these capacitors involves vibration frequencies above 55 Hz.

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A.3.2.2.10 Increased reliability. Failure rate is a function of temperature, applied voltage, and circuit impedance. Increased reliability may be obtained by derating the temperature and applied voltage and increasing circuit impedances.

DC leakage current increases when either voltage or temperature is increased; the rate of increase is greater at the higher values of voltage and temperature. A point can be reached where the dc leakage current will avalanche and attain proportions that will permanently damage the capacitor. Consequently, capacitors should never be operated above their rated temperature and rated voltage for that temperature.

By increasing the circuit impedance, the leakage current is reduced. In life testing the solid tantalum capacitor, the capacitance and dissipation factor are very stable over long periods of time and hence are not a suitable measure of deterioration. Leakage current variation is a better indicator of capacitor condition. In the life test in MIL-PRF-39003, a maximum impedance of 3 ohms is allowed. It is recommended that a minimum circuit impedance of 3 ohms per applied volt (1.5 ohms per volt for CSR21) be utilized to attain improved reliability.

A.3.2.3 MIL-PRF-39006, CLR, Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, Established Reliability.

A.3.2.3.1 Use. These capacitors are limited to applications at 125 Vdc or below. Their primary use is in low voltage power supply filtering circuits but also may be used in coupling and bypassing, circuit isolation, tuning, timing, power factor correction and phase shifting applications. Their low leakage current (lowest of all the tantalum types) is not appreciable below +85°C; and at ordinary operating temperatures is comparable to good quality paper capacitors, yet they are much smaller in size. These capacitors have the highest capacitance per size, or volumetric efficiency, of all capacitor dielectrics.

A.3.2.3.2 Construction: These capacitors consist of a sintered-pellet, acting as the anode, which is electrochemically treated to form a layer of tantalum oxide dielectric. The electrolyte is a liquid, normally a sulfuric acid solution, which establishes contact with all of the complex surfaces of the anodized pellet. The case, of which the inner surface may be covered with anodized tantalum powder, or some other material, becomes the cathode. To prevent the loss of the liquid electrolyte, the case is normally grooved with a seal ring against a Teflon™ top gasket. The top of the case is then closed using a hermetic glass-to-metal seal.

A.3.2.3.3 Voltage derating. When properly derated, these units may be operated over a temperature range of -55°C to +125°C. The derated voltage at +125°C is approximately 66 percent of the full rated voltage.

A.3.2.3.4 Reverse voltage. Styles CLR79, CLR81, CLR90 and CLR91 capacitors are for dc application only; however, they will withstand up to 3 volts of reverse bias. Other CLR styles should not be reverse biased and have substantially lower or NO reverse capability.

A.3.2.3.5 AC ripple current. The RMS ripple current ratings at 401K Hz, 85°C ambient and rated voltage are included in the data sheets. Capacitors must be biased such that the negative peak does not exceed -3 volts, and the DC voltage plus the positive peak does not exceed the rated working voltage of the capacitor. Other CLR styles have substantially lower or NO ripple current capability.

A.3.2.3.6 Series and parallel applications. Whenever tantalum capacitors with unequal values of dc leakage or capacitance are connected in series for higher voltage operation, it may be desirable to place a resistor in parallel across each unit. Unless a shunt resistor is used, the dc rated voltage can be exceeded on the capacitor in the series network with the highest impedance at the active circuit frequencies.

To obtain a higher capacitance than can be obtained from a single capacitor, a number of units may be connected in parallel. However, the sum of the peak ripple and the applied dc voltage should not exceed the dc rated voltage. The connecting leads of the parallel network should be large enough to carry the combined currents without reducing the effective capacitance due to series lead resistance.

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A.3.2.3.7 Reliability. Capacitors covered by this specification have reliability established for FRL ranging from 1.0 percent to 0.001 percent per 1,000 hours in accordance with MIL-STD-690. These FRLs are established at a 60-percent confidence level and are maintained at a 10-percent produce's risk. Ongoing testing to maintain these FRLs is conducted at full rated voltage and +85°C. Increased reliability may be obtained by derating the temperature and applied voltage, and increasing circuit impedance.

A.3.2.3.8 Mounting. Supplementary mounting means should be used where the application of these capacitors involves vibration frequencies above 55 Hz.

A.3.2.4 MIL-PRF-39018, CUR and CU, Capacitor, Fixed, Electrolytic (Aluminum Oxide), Established Reliability and Non-Established Reliability.

A.3.2.4.1 Use. These aluminum oxide electrolytic capacitors are intended for use in filter and bypass applications where large capacitance values are required in small cases and where excess of capacitance over the nominal value can be tolerated. Aluminum electrolytic capacitors provide the smallest volume, mass, and cost per microfarad of any type of capacitor with the exception of the tantalum electrolytic capacitor. For polarized capacitors, the applied ac peak voltage should never exceed the applied dc voltage; the sum of the applied ac peak and dc voltages should never exceed the dc rated voltage.

These capacitors are not recommended for airborne equipment applications since they should not be subjected to low barometric pressure and low temperatures at high altitudes. These aluminum electrolytic capacitors can be derated only for a short period since derating for any length of time may result in the necessity for re-forming. Even though they have vents designed to open at dangerous pressures, explosions can occur because of gas pressure or a spark ignition of free oxygen and hydrogen liberated at the electrodes. Provisions should be made to protect surrounding parts.

These capacitors are generally used where low frequency, pulsating, dc signal components are to be filtered out, and as cathode by-pass capacitors in self-biasing circuits. These capacitors are designed for applications where accuracy of capacitance is relatively unimportant.

As a rule, for selection of emitter by-pass capacitors, a ratio of bias resistance to by-pass reactance of about 10 to 1 is allowed. Ratios up to 20 to 1 may be used in high-fidelity-amplifier work or where space and economical considerations permit. Electrolytic capacitors provide the equipment designer with an unusually lightweight unit of high capacitance in a compact container.

ER capacitors covered by this specification have reliability established on the basis of life tests performed at rated voltage at +85°C for FRLs ranging from 1.0 percent to 0.001 percent per 1,000 hours in accordance with MIL-STD-690. These FRLs are established at a 60-percent confidence level and are maintained at a 10-percent producer's risk.

A.3.2.4.2 Construction. The capacitor consists of aluminum foil rolled onto a porous spacer. The foil is approximately .003 to .005 inch (0.08 to 0.13 mm) thick. The spacer is impregnated with an electrolyte and separates the anode and cathode. The electrolyte is usually an aqueous solution of ammonium borate, boric acid, and glycol. The metal cases are provided with an insulating sleeve which has an insulation resistance of at least 100 megohms.

It should be noted that the insulation resistance refers to the sleeve and not to the resistance between the terminals and the case. For proper performance, the case of all styles should be considered to be at the same potential as the negative terminal.

A.3.2.4.3 Voltage rating. The thickness of the oxide film which is formed both initially on the foil and during the forming operations on the completed capacitor determines the maximum peak or surge voltage which may be applied. For maximum reliability and long life, the dc working voltage should not be more than approximately 80 percent of full rating so that surges can be kept within the full-rated working voltage. The time of surge-voltage application should not be more than 30 seconds every 10 minutes.

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A.3.2.4.4 Derating. Style CU15 capacitors may be voltage derated in order to operate at temperatures up to +125°C. The percent of derating varies from approximately 20 to 33 percent depending on the particular voltage rating involved.

A.3.2.4.5 Seal. Even though these capacitors have vents designed to open at dangerous pressures, explosions can occur because of gas pressure or a spark ignition of free oxygen and hydrogen liberated at the electrodes. Provisions should be made to protect surrounding parts.

A.3.2.4.6 Polarization. Nonpolarized capacitors should be used in applications where reversal of potential occurs. Polarized capacitors should be used only in dc circuits with polarity properly observed. If ac components are present, the sum of the peak ac voltage plus the applied dc voltage must not exceed the dc rating. The peak ac value should also be less than the applied dc voltage so that polarity may be maintained, even on negative peaks, to avoid overheating and damage.

A.3.2.4.7 Storage shelf life and reforming. Storage shelf life and reforming guidelines can be found in MIL-HDBK-1131.

A.3.2.5 MIL-PRF-49137, CX, Capacitor, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Molded, Conformal Coated and Metal Cased with Plastic End-Fill, Nonhermetically Sealed.

A.3.2.5.1 Use. These tantalum electrolytic capacitors covered by this specification are intended mainly for use in filter, bypass, coupling, blocking, and other low-voltage applications (such as transistor circuits) where stability, size, weight, and shelf life are important factors. These capacitors are intended to be used only where supplemental moisture protection is provided or for noncritical applications where hermetic moisture protection is not required.

A.3.2.6 MIL-PRF-55365, CWR, Capacitor, Fixed, Electrolytic (Tantalum), Chip, Established Reliability, Nonestablished Reliability, and High Reliability.

A.3.2.6.1 Use. These tantalum chip capacitors are primarily intended for use in thick and thin film hybrid circuits or surface mount applications for filter, bypass, coupling, and other applications where the alternating current (ac) component is small compared to the direct current (dc) rated voltage and where supplemental moisture protection is available. Long term exposure to humid conditions may affect the equivalent series resistance of these capacitors and possibly result in damage during solder reflow. The ER capacitors have reliability ratings established in exponential distribution and Weibull distribution.

The reliability ratings are determined on the basis of life tests performed at specified voltage at +85°C. The exponential failure rates are 1.0 percent per 1,000 hours to 0.001 percent per 1,000 hours. These exponential failure rates are established at a 60-percent confidence level and are maintained at a 10-percent producer's risk. The Weibull failure rates are 0.1 percent per 1,000 hours to 0.0001 percent per 1,000 hours at a 90-percent confidence level.

The high reliability capacitors (T level) are intended for space, missile, and other high reliability applications and require the following: Weibull grading to the C FRL or lower, surge current option C, destructive physical analysis, 100 percent radiographic inspection, 100 percent visual inspection, and DC leakage and equivalent series resistance measurements within +3 standard deviations of the lot mean. Inspection of product for delivery for T level capacitors consists of group A and subgroups I, II, and III of group C. These additional tests and inspections are expected to result in a lower FRL than the initial Weibull graded lot.

A.3.2.6.2 Construction. A porous tantalum slab serves as the anode. The surfaces of the anode are electrochemically converted to an oxide of tantalum which serves as the dielectric. These surfaces are coated with an oxide semiconductor which is the working electrolyte in solid form.

A.3.2.6.3 Voltage derating. The derated voltage at +125°C is approximately 66 percent of the full rated voltage. While this capacitor type is designed for full rated voltage between -55°C and +85°C, historical application has also included a recommended minimum de-rating of 50% rated voltage.

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A.3.2.6.4 Surge current. Surge current testing is performed to verify the ability of capacitors to withstand high current peaks at rapid turn-on/off conditions in specific applications. It has been found to be very effective at screening out this potential failure mechanism. The surge current testing is optional. The specific test and options are detailed in MIL-PRF-55365.

A.3.2.6.5 Mounting. These capacitors are designed for mounting by reflow solder or conductive epoxy on circuit substrates.

A.3.2.6.6 Moisture and handling sensitivity. These capacitors may be sensitive to moisture induced stress if exposed to humidity for extended periods of time. J-STD-020 (Moisture/Reflow Sensitivity Classification for Nonhermetic Surface Mount Devices) provides a procedure to identify the moisture sensitivity level rating for non-hermetic surface mount devices so that they can be properly packaged, stored, and handled. The MSL ratings for the capacitors qualified to this specification are based on the tin/lead reflow profile specified in J-STD-020. If humidity exposure exceeds J-STD-020 limits, users may consult J-STD-033 (Handling, Packing, Shipping and Use of Moisture/Reflow Sensitive Surface Mount Devices) for general bake out and handling procedures; however, individual capacitor manufacturers may have more specific recommendations, especially for those capacitors packaged in tape and reel. Passing the criteria specified in J-STD-020 is not sufficient by itself to provide assurance of long-term reliability.

A.3.2.6.6.1 Solder reflow. Tantalum chip capacitors may become susceptible to damage during solder reflow due to moisture absorption. Vapor pressure of moisture inside a non-hermetic package increases greatly when exposed to the high temperature of solder reflow, potentially resulting in internal cracking or damage to the capacitor. In the most severe cases, the internal pressure can cause external package cracks, commonly referred to as the “popcorn” phenomenon.

A.3.2.6.6.2 ESR degradation. The ESR may increase due to moisture absorption. Moisture can penetrate the non-hermetic package and be absorbed by the cathode elements. This absorption can reduce conductivity of the contact points which increases the resistance. This process may be reversible by eliminating the moisture within the capacitor by performing a bake out drying procedure.

A.3.3 General characteristics of glass capacitors.

A.3.3.1 MIL-PRF-14409, PC, Capacitor, Variable (Piston Type, Tubular Trimmer).

A.3.3.1.1 Use. These capacitors are small-sized, tubular-trimmer, variable capacitors designed for use where fine tuning adjustments are periodically required during the life of the equipment. Normally they are used for trimming and coupling in such circuits as intermediate frequency, radio-frequency, oscillator, phase shifter, and discriminator stages. Because of their low mass, these units are relatively stable against shock and vibration which tend to cause changes in capacitance. Capacitance change versus rotation is linear within ±10 percent. Backlash is virtually nonexistent except on styles PC39 and PC43 which have a maximum backlash of 2 percent.

A.3.3.1.2 Construction. Styles PC18, PC46, PC19, PC47, PC25, PC26, PC30, and PC32 capacitors are constructed of a series of concentric circular metal bands which interleaf and are variable by adjustment of the related depth of the interface. All other style capacitors are constructed of glass, quartz, sapphire, or alumina dielectric cylinders and metal tuning pistons. A portion of the cylinder is plated with metal to form the stator and the metal piston, controlled by a tuning screw, acts as the rotor for these variable capacitors. The overlap of the stator and rotor determines the capacitance. The self-contained piston within the dielectric cylinder functions as a low inductance coaxial assembly.

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A.3.4 General characteristics of mica capacitors.

A.3.4.1 MIL-PRF-39001, CMR, Capacitors, Fixed, Mica Dielectric, Established Reliability and Nonestablished Reliability.

A.3.4.1.1 Use. These mica capacitors are intended for use in equipment where a known order of reliability is required. The ER capacitors have a FRL ranging from 1.0 percent per 1,000 hours to 0.001 percent per 1,000 hours. The FRL is established at a 90 percent confidence level. An acceleration factor of 25:1 has been used to relate the life test at 150 percent of rated voltage at rated temperature to rated voltage at rated temperature.

A.3.4.1.2 Construction. Muscovite mica is the most commonly used material. It has a dielectric constant between 6.5 and 8.5 and can be split into thin sheets; it is nonporous and does not readily absorb moisture. Protection from moisture is provided to obtain high-capacitance stability and low losses. The two techniques used to form the capacitors covered in this section are by stacking the mica sheets through the silvered-mica process or by the use of tin-lead foil to separate the mica sheets. The molded units referenced in this section are fixed terminal capacitors. Terminals are attached to the mica stacks by the use of pressure clips which have been solder-coated for maximum mechanical strength. The molded case is made of a polyester material which also exhibits high insulation resistance and high resistance to moisture absorption and transmission. The molded case also imparts rigidity to the capacitor in the event the capacitor is subjected to vibration or shock.

A.3.5 General characteristics of paper and plastic capacitors.

A.3.5.1 MIL-PRF-11693, CZR and CZ, Capacitors, Feed Through, Radio-Interference Reduction, DC (Hermetically Sealed in Metal Cases), Established and Non-Established Reliability.

A.3.5.1.1 Use. These capacitors are primarily designed for use in broadband, radio-interference suppression applications. Capacitors meeting the established reliability requirements specified herein have a maximum failure rate of 1%/1000 hours. This failure rate is established with a 90% confidence limit based on the life test parameters specified and are maintained at a 10-percent producer's risk. An acceleration factor of 5:1 has been used to relate the life test data obtained at 140 percent of rated dc voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature.

A.3.5.2 MIL-PRF-19978, CQR and CQ, Capacitors, Fixed, Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metal, Ceramic or Glass Cases), Established and Non-Established Reliability.

A.3.5.2.1 Use. These capacitors are designed for use in circuit applications requiring high insulation resistance, low dielectric absorption, or low loss factor over wide temperature ranges, and where the ac component of the impressed voltage is small with respect to the dc voltage rating. These capacitors may be used where an ac component is present provided that: (1) the sum of the dc voltage and the peak ac voltage does not exceed the dc voltage rating or (2) the peak ac voltage does not exceed 20 percent of the dc voltage rating at 60 Hz, 15 percent at 120 Hz; or 1 percent at 10,000 Hz. Where heavy transient or pulse currents are encountered, the requirements of MIL-PRF-19978 are not sufficient to guarantee satisfactory performance, and due allowance must therefore be made in the selection of a capacitor.

Polyethylene terephthalate dielectric capacitors (characteristic M). Capacitors of polyethylene terephthalate dielectric are intended for use in high temperature applications similar to those served by hermetically sealed paper capacitors, but where higher insulation resistance at the upper temperature limits is required.

Paper and polyethylene terephthalate dielectric capacitors (characteristics E, F, G, and K). Capacitors of paper and polyethylene terephthalate dielectric are intended for applications where small case sizes and high temperature operation are required.

Polytetrafluoroethylene dielectric capacitors (characteristic T). Capacitors of polytetrafluoroethylene dielectric are intended for high temperature applications where high insulation resistance, small capacitance change, and low dielectric absorption are required. These capacitors exhibit excellent insulation resistance values at high temperatures.

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Polycarbonate (PC) (characteristic Q). These capacitors are intended for applications where minimum capacitance changes with temperature is required; they are especially suitable for use in tuned and precision timing circuits. Capacitor grade polycarbonate film is in limited supply since the sole source stopped producing it in 2000. Polyphenylene sulfide (PPS) capacitors are currently the closest alternative available for polycarbonate capacitors. Insulation resistance, dissipation factor and capacitance change with temperature are slightly different for PPS capacitors than PC capacitors.

Polyphenelyene sulfide capacitors (PPS) These capacitors offer very similar DC properties to polycarbonate capacitors. The ac properties (insulation resistance, dissipation factor and capacitance change with temperature) are slightly different. PPS capacitors are good choices for many applications. When used as replacements for polycarbonate capacitors, they need to be evaluated for each application to assure that they meet the requirements.

ER capacitors have FRLs ranging from 1.0 percent to 0.001 percent per 1,000 hours. These FRLs are established at a 90-percent confidence level and maintained at a 10-percent producer's risk and are based on life tests performed at maximum rated voltage and maximum rated temperature. An acceleration factor of 5:1 has been used to relate the life test data obtained at 140 percent of rated dc voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature. This specification also covers removable mounting retainers for use with applicable capacitors.

A.3.5.2.2 Construction. The present drive towards miniaturization, closer electrical tolerances, and higher operating temperatures is being met by the use of thin plastic-film dielectrics in the construction of capacitors. The greatest advantage of plastic-film dielectrics over natural dielectrics (such as paper and mica) is that the plastic film is a synthetic that can be made to meet specific requirements (such as thickness of dielectric and high heat-resistance). Many plastic-film capacitors are not impregnated but are wound and encased "dry". Plastic dielectric capacitors have insulation resistance values far in excess of those for paper capacitors and, since they are nonabsorbent, their moisture characteristics are superior to those of mica.

There are several types of plastic films available for use as a capacitor dielectric. They may be used individually or in a combination with other films and with paper in order to obtain the compromised advantages of the specific electrical characteristics of each individual film. The more common films include polyethylene terephthalate and polycarbonate. When properly applied, plastic dielectric films lead to the solution of many special capacitor problems. Capacitors using polyethylene terephthalate film as the dielectric are perhaps the most common of the plastic film types on the market today. Some manufactures use only one sheet of plastic film for those with low voltage ratings whereas at least two sheets of paper are used in conventional paper types. The principal advantage of polyethylene terephthalate dielectric capacitors is the high order of insulation resistance values available over the dielectric temperature range of -55°C to +125°C; however, for military applications, the high temperature limit is +85°C. Polyethylene terephthalate dielectric capacitors have an insulation resistance that is normally about 100,000 megohms per microfarad at room temperature and about 25,000 megohms per microfarad at +85°C. These insulation resistance values decrease considerably when polyethylene terephthalate dielectric capacitors are impregnated. However, a higher volt per mil rating is possible by impregnation and the possibility of corona and catastrophic failures due to pinholes in the dielectric are minimized.

A.3.5.2.3 Barometric pressure (flashover) for metal-cased tubular capacitors. The dc voltage that may be applied to metal-cased tubular capacitors at altitudes other than 80,000 feet may be obtained from a figure in MIL-PRF-19978, except that the dc voltage ratings must not be exceeded.

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A.3.5.3 MIL-PRF-39022, CHR, Capacitors, Fixed, Metallized, Paper-Plastic Film, or Plastic Film Dielectric, Direct and Alternating Current, (Hermetically Sealed in Metal or Ceramic Cases), Established Reliability.

A.3.5.3.1 Use. These capacitors are primarily intended for use in power supply filter circuits, by-pass applications, and other applications where the alternating current (ac) component of voltage is small with respect to the applied dc voltage and where occasional periods of low-insulation resistance and momentary breakdowns can be tolerated.

The FRLs are established at a 90 percent confidence level and maintained at a 10 percent producer's risk and are based on life tests performed at maximum rated voltage and maximum rated temperature. Unless otherwise specified an acceleration factor of 5:1 has been used to relate life test data obtained at 140 percent of rated voltage at maximum rated temperature, to rated voltage at rated temperature.

A.3.5.3.2 Construction. The construction of metallized plastic capacitors differs from conventional plastic capacitors in that instead of having separate layers of metal foil (capacitor plates) and plastic dielectric, the metal comprising the capacitor plates is imposed directly on one side of the plastic dielectric by means of a metallizing process. This technique results in an overall size reduction for metallized plastic capacitors when compared to conventional plastic-foil capacitor types of equal ratings. This space saving is the outstanding feature of the metallized plastic capacitor. A 200-volt metallized plastic capacitor has 0.75 the volume of the conventional plastic construction; at 600 volts, this ratio increases to 0.8; above 600 volts, the metallized plastic capacitor provides no size advantage.

Another advantage resulting from the metallizing technique is that the capacitors are self-healing. The metallic film imposed on the plastic is very thin and if breakdown occurs, a tiny area of the thin film surrounding the breakdown point burns away, leaving the capacitor operable, but with a slightly reduced capacitance. In the conventional plastic-foil type (where the foil is thicker), sustained conduction can occur on a breakdown causing a large area of the plastic surrounding the breakdown to be carbonized resulting in a permanent short-circuit.

The breakdown of the metallized plastic capacitor can be either of two types; i.e., (1) a complete breakdown lasting for only a moment (momentary breakdown) or (2) a sharp reduction in insulation resistance lasting for an extended period of time, but eventually returning to normal (period of low insulation). The general characteristics of the metallized plastic type, aside from the breakdowns, are similar to the conventional plastic type except for a significantly lower insulation resistance, approximately in the order of 10 to 1.

A.3.5.3.3 Prevention of corona. All metal parts, fittings, conductors, and attachments which operate at higher potential than other adjacent parts of the housing, should be carefully finished in order to insure that all sharp corners and edges are removed to minimize the possibility of corona discharge. Parts, from which the removal of sharp corners and edges would be impractical, such as conductors, should be spaced in such a manner as to prevent harmful corona discharges.

A.3.5.3.4 Mounting. Capacitors with dimension L or D of 1.375 or .672 inches (34.93 or 17.07 mm), respectively, and greater, should not be supported by their leads. These capacitors should be provided with a supplementary means for mounting, such as a tangential bracket.

A.3.5.4 MIL-PRF-55514, CFR, Capacitors, Fixed, Plastic (or Metallized Plastic) Dielectric, DC or DC-AC, in Nonmetal Cases, Non-Established and Established Reliability.

A.3.5.4.1 Use. These capacitors are designed for use in circuit applications requiring high insulation resistance, low dielectric absorption, or low loss factor over wide temperature ranges, and where the ac component of the impressed voltage is small with respect to the dc voltage rating.

NOTE: These capacitors may be used where an ac component is present provided that (1) the sum of the dc voltage and the peak ac voltage does not exceed the dc voltage rating or (2) the peak ac voltage does not exceed 20 percent of the dc voltage rating at 60 Hz; 15 percent at 120 Hz; or 1 percent at 10,000 Hz. Where heavy transient or pulse currents are encountered, the requirements of MIL-PRF-55514 are not sufficient to guarantee satisfactory performance, and due allowance must therefore be made in the selection of a capacitor.

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Capacitors acquired under provisions of this specification will experience failure at a rate depending almost exclusively upon the manner in which the capacitors are used. For military applications, in view of their limited long term moisture resistant characteristics, these capacitors should be used in potted or encapsulated circuit packaging systems. In any case, the acquiring activity should review the specific application. Where the capacitors are not incorporated in an encapsulated or potted circuit package, hermetically sealed foil capacitors of MIL-PRF-19978 or hermetically sealed metallized capacitors of MIL-PRF-39022 or MIL-PRF-83421 are recommended. In addition, the life of the capacitors specified herein is primarily dependent upon the operating temperature and applied voltage. They should not be used above rated voltage or temperature.

Polyethylene terephthalate. These capacitors are intended for high-temperature applications similar to those served by hermetically-sealed paper capacitors, but where high insulation resistance at the upper temperature limits is required.

Polycarbonate (PC). These capacitors are intended for applications where minimum capacitance changes with temperature is required; they are especially suitable for use in tuned and precision timing circuits. Capacitor grade polycarbonate film is in limited supply since the sole source stopped producing it in 2000. Polyphenylene sulfide (PPS) capacitors are currently the closest alternative available for polycarbonate capacitors. Insulation resistance, dissipation factor and capacitance change with temperature are slightly different for PPS capacitors than PC capacitors.

Polyphenelyene sulfide capacitors (PPS) These capacitors offer very similar DC properties to polycarbonate capacitors. The ac properties (insulation resistance, dissipation factor and capacitance change with temperature) are slightly different. PPS capacitors are good choices for many applications. When used as replacements for polycarbonate capacitors, they need to be evaluated for each application to assure that they meet the requirements.

ER capacitors covered by this specification have FRLs ranging from 1.0 percent to 0.001 percent per 1,000 hours. These FRLs are established at a 90-percent confidence level and maintained at a 10-percent producer's risk and based on life tests performed at +85°C. An acceleration factor of 5:1 has been used to relate life test data obtained at 125 percent, or 140 percent of rated voltage at +85°C, to rated voltage at +85°C. The product levels are based on catastrophic failures and failures occurring outside the degradation limits.

A.3.5.4.2 Construction.

Plastic film. The present drive towards miniaturization, closer electrical tolerances, and higher operating temperatures is being met by the use of thin plastic-film dielectrics in the construction of capacitors. The greatest advantage of plastic-film dielectrics over natural dielectrics (such as paper and mica) is that the plastic film is a synthetic that can be made to meet specific requirements (such as thickness of dielectric and high heat resistance). Many plastic-film capacitors are not impregnated but are wound and encased "dry". Plastic dielectric capacitors have insulation resistance values far in excess of those for paper capacitors and since they are nonabsorbent, their moisture characteristics are superior to those of mica.

There are several types of plastic films available for use as a capacitor dielectric. They may be used individually or in a combination with other films in order to obtain the compromised advantages of the specific electrical characteristics of each individual film. The more common films include polyethylene terephthalate and polycarbonate. When properly applied, plastic dielectric films lead to the solution of many special capacitor problems.

Metallized plastic. The construction of metallized plastic capacitors differs from conventional plastic capacitors in that instead of having separate layers of metal foil (capacitor plates) and plastic dielectric, the metal comprising the capacitor plates is imposed directly on one side of the plastic dielectric by means of a metallizing process. This technique results in an overall size reduction for metallized plastic capacitors when compared to conventional plastic-foil capacitor types of equal ratings. This space saving is the outstanding feature of the metallized plastic capacitor.

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Another advantage resulting from the metallizing technique is that the capacitors are self-healing. The metallic film imposed on the plastic is very thin and if breakdown occurs, a tiny area of the thin film surrounding the breakdown point burns away, leaving the capacitor operable, but with a slightly reduced capacitance. In the conventional plastic-foil type (where the foil is thicker), sustained conduction can occur on a breakdown causing a large area of the plastic surrounding the breakdown to be carbonized resulting in a permanent short-circuit.

The breakdown of the metallized plastic capacitor can be either of two types; i.e., (1) a complete breakdown lasting for only a moment (momentary breakdown) or (2) a sharp reduction in insulation resistance lasting for an extended period of time, but eventually returning to normal (period of low insulation). The general characteristics of the metallized plastic type, aside from the breakdowns, are similar to the conventional plastic type except for a significantly lower insulation resistance, approximately in the order of 10 to 1.

A.3.5.4.3 Mounting. Capacitors covered by this specification should be mounted by a bracket or clamp, or they should be potted when vibration or shock are likely to be encountered in service. When a bracket or clamp is used, care should be taken to assure that the capacitor body is not deformed.

A.3.5.5 MIL-PRF-83421, CRH, Capacitors, Fixed, Metallized, Plastic Film Dielectric, (DC, AC, or DC and AC), Hermetically Sealed in Metal or Ceramic Cases, Established Reliability.

A.3.5.5.1 Use. These metallized plastic film capacitors are primarily intended for use in power supply filter circuits, by-pass applications, and other applications where the ac component of voltage is known or a significant factor. Styles covered by this specification may be used for applications requiring a tight capacitance tolerance, excellent capacitance stability, very high-insulation resistance, and low-loss factors where the ac component of voltage is large with respect to the applied dc voltage.

Polycarbonate (PC). The dielectric of MIL-PRF-83421/1 capacitors is polycarbonate film. Capacitor grade polycarbonate film is in limited supply since the sole source stopped producing it in 2000. Polyphenylene sulfide (PPS) (MIL-PRF-83421/6) capacitors are currently the closest alternative available for polycarbonate capacitors. Insulation resistance, dissipation factor and capacitance change with temperature are slightly different for PPS capacitors than PC capacitors.

Polyphenelyene sulfide capacitors (PPS) The dielectric of MIL-PRF-83421/6 PPS film. These capacitors offer very similar DC properties to polycarbonate capacitors. The ac properties (insulation resistance, dissipation factor and capacitance change with temperature) are slightly different. PPS capacitors are good choices for many applications. When used as replacements for polycarbonate capacitors, they need to be evaluated for each application to assure that they meet the requirements.

Capacitors covered by this specification have FRLs established at a 90 percent confidence level and maintained at a 10 percent producer's risk and, unless otherwise specified, are based on life tests performed at maximum rated voltage at maximum rated temperature. Unless otherwise specified, an acceleration factor of 5:1 has been used to relate life-test data obtained at 140 percent of rated voltage at maximum rated temperature, to rated voltage at rated temperature.

A.3.5.5.2 Construction. Metallized plastic film capacitors differ from plastic foil types which have separate layers of metal foil (capacitor plates) and plastic dielectric. The metal comprising the metallized capacitor plates is a thin conductive coating on one side of the plastic dielectric by means of a metallizing process. This technique results in an overall size reduction for metallized plastic capacitors when compared to plastic foil capacitors of equal voltage rating and capacitance value. Typically, a 1 μF, 50 volts dc metallized polycarbonate capacitor will occupy approximately one-third the volume of a similar polycarbonate foil capacitor.

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Another advantage resulting from the metallizing technique is that the capacitors are self-healing. Generally, the voltage breakdown occurs through a small hole or thin spot in the dielectric with the fault current melting away the conductive metal coating adjacent to the fault area. After clearing, the capacitors will continue to operate normally with the possibility of reduced insulation resistance, increased dielectric absorption and no significant change to capacitance value or dissipation factor. Clearing will occur only if there is sufficient energy available from the circuit and/or stored in the capacitor. Minimum stored energy in the range of 100 to 500 microjoules is recommended to insure clearing. Applications for these capacitors should be limited to circuits that will provide sufficient energy to insure clearing and are insensitive to momentary breakdowns (clearing actions). In the conventional plastic-foil types (where the foil is thicker), sustained conduction can occur on a breakdown causing a large area of the plastic surrounding the breakdowns to be carbonized resulting in a permanent short-circuit.

The breakdown of the metallized plastic capacitor can be either of two types; i.e., (1) a complete breakdown lasting for only a moment (momentary breakdown) or (2) a sharp reduction in insulation resistance lasting for an extended period of time, but eventually returning to normal (period of low insulation). The general characteristics of the metallized plastic type, aside from the breakdowns, are similar to the conventional plastic type except for a significantly lower insulation resistance, approximately in the order of 10 to 1.

A.3.5.5.3 Prevention of corona. All metal parts, fittings, conductors, and attachments which operate at higher potential than other adjacent parts of the housing, should be carefully finished in order to insure that all sharp corners and edges are removed to minimize the possibility of corona discharge. Parts, from which the removal of sharp corners and edges would be impractical, such as conductors, should be spaced in such a manner as to prevent harmful corona discharges.

A.3.5.5.4 Mounting. Capacitors with dimension L or D of 1.375 or .670 inches (34.93 or 17.20 mm), respectively, and greater, should not be supported by their leads. These capacitors should be provided with a supplementary means for mounting, such as a tangential bracket.

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TABLE A-II. Capacitor selection guidance table.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-20 (Fixed, ceramic, temperature compensating established and non-established reliability)	27	CCR75	Tubular, axial lead (0.160 X 0.090)	.1 pF to 75 pF (200V) 82 pF to 240 pF (100V) 270 pF to 680 pF (50V)
	28	CC76/CCR76	Tubular, axial lead (0.250 X 0.090)	82 pF to 130 pF (200V) 270 pF to 680 pF (100V) 750 pF to 1,000 pF (50V)
	29	CCR77	Tubular, axial lead (0.390 X 0.140)	150 pF to 680 pF (200V) 750 pF to 2,200 pF (100V) 2,400 pF to 5,600 pF (50V)
	30	CC78/CCR78	Tubular, axial lead (0.500 X 0.250)	820 pF to 3,300 pF (200V) 3,900 pF to 12,000 pF (100V) 15,000 pF to 27,000 pF (50V)
	31	CC79/ CCR79	Tubular, axial lead (0.690 X 0.350)	3,900 pF to 10,000 pF (200V) 15,000 pF to 39,000 pF (100V) 47,000 pF to 82,000 pF (50V)
	35	CCR05/CCR09	Molded, radial lead (0.190 X 0.190)	1 pF to 330 pF (200V) 360 pF to 1,800 pF (100V) 2,000 pF to 3,300 pF (50V)
	36	CCR06	Rectangular, radial lead (0.290 X 0.290)	360 pF to 1,800 pF (200V) 2,000 pF to 4,700 pF (100V) 5,100 pF to 18,000 pF (50V)
	37	CC07/CCR07	Rectangular, radial lead (0.480 X 0.480)	2,200 pF to 4,700 pF (200V) 5,600 pF to 12,000 pF (100V) 15,000 pF to 100,000 pF (50V)
	38	CC08/CCR08	Rectangular, radial lead (0.480 X 0.480)	3,900 pF to 4,700 pF (200V) 15,000 pF to 18,000 pF (100V) 56,000 pF to 68,000 pF (50V)
MIL-PRF-81 (Variable, ceramic)	1	CV11	Variable	1.5-7.0 pF to 7.0-45.0 pF (500V)
	4	CV31/CV32	Variable	2.0-8.0 pF to 8.0-25.0 pF (350V) 3.0-15.0 pF to 15.0-60.0 pF (200V)
	6	CV35/CV36	Variable	1.0-3.0 pF to 7.0-40.0 pF (100V) 7.0-40.0 pF (25V)
MIL-PRF-123 (Ceramic, temperature stable and general purpose, high reliability)	1	CKS05	Molded, radial lead	4.7 pF to 4,700 pF (100V) 270 pF to 10,000 pF (50V)
	2	CKS06	Molded, radial lead	270 pF to 100,000 pF (100V) 2,700 pF to 1,000,000 pF (50V)
	3	CKS07	Molded, radial lead	2,700 pF to 470,000 pF (100V) 11,000 pF to 1,000,000 pF (50V)
	4	CKS11	Molded, axial lead	1.1 pF to 75 pF (200V) 4.7 pF to 1,000 pF (100V) 110 pF to 4,700 pF (50V)
	5	CKS12	Molded, axial lead	82 pF to 130 pF (200V) 110 pF to 4,700 pF (100V) 240 pF to 10,000 pF (50V)
	6	CKS14	Molded, axial lead	150 pF to 680 pF (200V) 240 pF to 10,000 pF (100V) 1,100 pF to 47,000 pF (50V)
	7	CKS15	Molded, axial lead	820 pF to 3,300 pF (200V) 1,100 pF to 100,000 pF (100V) 2,400 pF to 180,000 pF (50V)
	8	CKS16	Molded, axial lead	3,900 pF to 10,000 pF (200V) 2,400 pF to 470,000 pF (100V) 11,000 pF to 1,000,000 pF (50V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-123 (Continued)	10	CKS51	Chip (0805)	1 pF to 4,700 pF (100V) 1 pF to 18,000 pF (50V)
	11	CKS52	Chip (1210)	300 pF to 2,200 pF (100V) 300 pF to 47,000 pF (50V)
	12	CKS53	Chip (1808)	300 pF to 3,300 pF (100V) 300 pF to 100,000 pF (50V)
	13	CKS54	Chip (2225)	1,100 pF to 1,000,000 pF (50V)
	16	CKS22	Molded, 2 pin DIP	1 pF to 820 pF (200V) 560 pF to 10,000 pF (100V) 2,700 pF to 100,000 pF (50V)
	17	CKS23	Molded, 2 pin DIP	560 pF to 10,000 pF (200V) 2,700 pF to 100,000 pF (100V) 4,700 pF to 220,000 pF (50V)
	18	CKS24	Molded, 2 pin DIP	120,000 pF to 150,000 pF (100V) 180,000 pF to 470,000 pF (50V)
	21	CKS55	Chip (1206)	1 pF to 15,000 pF (100V) 1,200 pF to 39,000 pF (50V)
	22	CKS56	Chip (1812)	1,200 pF to 56,000 pF (100V) 5,100 pF to 180,000 pF (50V)
	23	CKS57	Chip (1825)	3,900 pF to 150,000 pF (100V) 11,000 pF to 470,000 pF (50V)
MIL-PRF-11693 (Feed through, radio-interference reduction)	7	CZ23, CZR23, CZ24, CZR24	Tubular	0.0047 µF to 0.47 µF (600V) 0.1 µF to 0.47 µF (400V) 0.047 µF to 1.0 µF (200V) 0.047 µF to 2.0 µF (100V)
MIL-PRF-14409 (Variable, piston type, tubular trimmer)	4	PC38	Piston type, tubular trimmer	0.6-1.8 pF (750V) 0.6-5.5 pF to 0.8-16 pF (1,250V)
	4	PC48	Piston type, tubular trimmer	0.8-5.5 pF (750V) 0.8-11.0 pF to 1.0-38.0 pF (1,250V)
	5	PC39	Piston type, tubular trimmer	1.0-16.0 pF to 1.0-120.0 pF (1000V)
	6	PC40	Piston type, tubular trimmer	0.6-1.8 pF to 0.8-16.0 pF (750V)
	6	PC50	Piston type, tubular trimmer	0.8-5.5 pF to 1.0-38.0 pF (750V)
	7	PC41	Piston type, tubular trimmer	0.6-1.8 pF to 0.8-16.0 pF (750V)
	7	PC51	Piston type, tubular trimmer	0.8-5.5 pF to 0.8-38.0 pF (750V)
	8	PC42	Piston type, tubular trimmer	0.6-5.5 pF to 0.8-16 pF (1,250V) 0.6-1.8 pF (750V)
	8	PC52	Piston type, tubular trimmer	0.8-11.0 pF to 0.8-38.0 pF (1,250V) 0.8-5.5 pF (750V)
	9	PC43	Piston type, tubular trimmer	1.0-16.0 pF to 1.0-120.0 pF (1,000V)
	12	PC25/PC31	Piston type, tubular trimmer	0.6-6.0 pF to 1.0-20.0 pF (250V)
	13	PC26	Piston type, tubular trimmer	0.6-6.0 pF to 1.0-30.0 pF (250V)
	14	PC27	Piston type, tubular trimmer	0.35-3.5 pF (250V)
	14	PC28	Piston type, tubular trimmer	0.35-3.5 pF to 0.5-5.0 pF (250V)
	14	PC29	Piston type, tubular trimmer	0.35-3.5 pF to 0.8-10.0 pF (250V)
	15	PC30	Piston type, tubular trimmer	0.6-3.5 pF to 0.6-5.0 pF (250V)
	15	PC32	Piston type, tubular trimmer	1.5-14.0 pF (250V)
	15	PC34	Piston type, tubular trimmer	0.8-10.0 pF (250V)
	16	PC21/PC22/PC23/ PC24	Piston type, tubular trimmer	0.3-1.2 pF to 0.8-8.0 pF (500V)
	17	PC17	Piston type, tubular trimmer	1.0-5.5 pF to 1.5-40.0 pF (250V)
	18	PC18/PC46	Piston type, tubular trimmer	1.0-10.0 pF (250V) 1.0-14.0 pF (125V)
	19	PC19/PC47	Piston type, tubular trimmer	1.0-10.0 pF (250V) 1.0-14.0 pF (125V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-19978 (Plastic or paper-plastic, hermetically sealed, established and nonestablished reliability)	1	CQ08/CQ09/CQ12 /CQ13	Tubular, axial lead, hermetically sealed, non-established reliability	0.0010 µF to 0.22 µF (600V) 0.0010 µF to 0.33 µF (400V) 0.0015 µF to .47 µF (200V) 0.039 µF to 1.0 µF (50V)
	2	CQ20	Tubular, axial lead, hermetically sealed in ceramic or glass, non-established reliability	0.00047 µF to 0.015 µF (15,000V) 0.00047 µF to 0.022 µF (12,500V) 0.00047 µF to 0.039 µF (10,000V) 0.00047 µF to 0.068 µF (7,500V) 0.00047 µF to 0.15 µF (5,000V) 0.001 µF to 0.33 µF (3,000V) 0.0010 µF to 0.68 µF (2,000V) 0.0010 µF to 1.0 µF (1,000V)
	3	CQ72	Rectangular, large can, hermetically sealed, non-established reliability	0.1 µF to 2.0 µF (12,500V, 10,000V) 0.1 µF to 2.0 µF (7,500V, 6,000V) 0.1 µF to 4.0 µF (5,000V, 4,000V) 0.1 µF to 6.0 µF (3,000V) 0.1 µF to 12.0 µF (2,500V) 0.25 µF to 15.0 µF (2,000V) 0.1 µF to 15.0 µF (1,500V, 1,000V) 0.25 µF to 15.0 µF (600V, 400V)
	9	CQR09	Tubular, insulated, axial lead, hermetically sealed, established reliability	0.001 µF to 0.47 µF (1,000V) 0.001 µF to 0.47 µF (600V) 0.0027 µF to 0.47 µF (400V) 0.0039 µF to 1.0 µF (200V)
	10	CQR12	Tubular, uninsulated, axial lead, tangential retainer, hermetically sealed, established reliability	0.001 µF to 0.47 µF (1,000V) 0.001 µF to 0.47 µF (600V) 0.0027 µF to 0.47 µF (400V) 0.0039 µF to 1.0 µF (200V)
	11	CQR13	Tubular, uninsulated, axial lea, threaded-stud retainer, hermetically sealed, established reliability	0.18 µF to 0.47 µF (1,000V) 0.001 µF to 0.47 µF (600V) 0.0027 µF to 0.47 µF (400V) 0.0039 µF to 1.0 µF (200V)
	13	CQR29	Tubular, insulated, axial lead, hermetically sealed, established reliability	0.001 µF to 0.22 µF (1,000V) 0.001 µF to 0.68 µF (600V) 0.001 µF to 1.0 µF (400V) 0.001 µF to 1.0 µF (200V) 1.2 µF to 6.8 µF (100V) 0.001 µF to 1.2 µF (50V) 1.5 µF to 10.0 µF (30V)
	14	CQR32	Tubular, uninsulated, axial lead, tangential retainer, hermetically sealed, established reliability	0.001 µF to 0.22 µF (1,000V) 0.001 µF to 0.68 µF (600V) 0.001 µF to 1.0 µF (400V) 0.001 µF to 1.0 µF (200V) 1.2 µF to 6.8 µF (100V) 0.001 µF to 1.2 µF (50V) 1.5 µF to 10.0 µF (30V)
	15	CQR33	Tubular, uninsulated, axial lead, threaded-stud retainer, hermetically sealed, established reliability	0.082 µF to 0.22 µF (1,000V) 0.001 µF to 0.68 µF (600V) 0.001 µF to 1.0 µF (400V) 0.001 µF to 1.0 µF (200V) 1.2 µF to 6.8 µF (100V) 0.001 µF to 1.2 µF (50V) 1.5 µF to 10.0 µF (30V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-19978 (Continued)	22	CQR44	Tubular, insulated, axial lead, hermetically sealed, established reliability	0.001 μF to 0.22 μF (600V) 0.0012 μF to 0.47 μF (400V) 0.001 μF to 0.68 μF (200V) 0.0022 μF to 1.0 μF (100V) 0.0039 μF to 1.0 μF (50V)
MIL-PRF-32535 (Ceramic, chip, temperature stable and general purpose, extended range, high reliability and standard reliability)	1	Size 0201	Chip (0201)	1,000 pF to 10,000 pF (10V) 1,000 pF to 10,000 pF (6.3V) 1,000 pF to 10,000 pF (4V)
	2	Size 0402	Chip (0402)	10 pF to 4,700 pF (100V) 10 pF to 27,000 pF (50V) 10 pF to 33,000 pF (25V) 10 pF to 33,000 pF (16V) 10 pF to 100,000 pF (10V) 10 pF to 100,000 pF (6.3V) 10 pF to 100,000 pF (4V)
	3	Size 0603	Chip (0603)	1.0 pF to 82 pF (200V) 1.0 pF to 18,000 pF (100V) 1.0 pF to 150,000 pF (50V) 1.0 pF to 180,000 pF (25V) 1.0 pF to 220,000 pF (16V) 1.0 pF to 220,000 pF (10V) 1.0 pF to 220,000 pF (6.3V) 1.0 pF to 220,000 pF (4V)
	4	Size 0805	Chip (0805)	1.0 pF to 470 pF (200V) 1.0 pF to 100,000 pF (100V) 1.0 pF to 470,000 pF (50V) 1.0 pF to 1,000,000 pF (25V) 1.0 pF to 1,000,000 pF (16V) 1.0 pF to 1,000,000 pF (10V) 1.0 pF to 1,000,000 pF (6.3V) 1.0 pF to 1,000,000 pF (4V)
	5	Size 1206	Chip (1206)	560 pF to 1,800 pF (200V) 560 pF to 390,000 pF (100V) 560 pF to 1,000,000 pF (50V) 560 pF to 2,200,000 pF (25V) 560 pF to 3,300,000 pF (16V) 560 pF to 3,300,000 pF (10V) 560 pF to 3,300,000 pF (6.3V) 560 pF to 3,300,000 pF (4V)
	6	Size 1210	Chip (1210)	1,200 pF to 3,300 pF (200V) 1,200 pF to 820,000 pF (100V) 1,200 pF to 1,000,000 pF (50V) 1,200 pF to 2,200,000 pF (25V) 1,200 pF to 3,300,000 pF (16V) 1,200 pF to 3,300,000 pF (10V) 1,200 pF to 3,300,000 pF (6.3V) 1,200 pF to 3,300,000 pF (4V)
	7	Size 1812	Chip (1812)	1,000 pF to 6,800 pF (200V) 1,000 pF to 2,200,000 pF (100V) 1,000 pF to 4,700,000 pF (50V) 1,000 pF to 8,200,000 pF (25V) 1,000 pF to 8,200,000 pF (16V) 1,000 pF to 8,200,000 pF (10V) 1,000 pF to 8,200,000 pF (6.3V) 1,000 pF to 8,200,000 pF (4V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-32535 (Continued)	8	Size 2220	Chip (2220)	1,000 pF to 27,000 pF (200V) 1,000 pF to 4,700,000 pF (100V) 1,000 pF to 10,000,000 pF (50V) 1,000 pF to 22,000,000 pF (25V) 1,000 pF to 22,000,000 pF (16V) 1,000 pF to 22,000,000 pF (10V) 1,000 pF to 22,000,000 pF (6.3V) 1,000 pF to 22,000,000 pF (4V)
	9	Size 0306	Chip (0306)	1,000 pF to 68,000 pF (25V) 1,000 pF to 100,000 pF (16V) 1,000 pF to 100,000 pF (10V) 1,000 pF to 100,000 pF (6.3V) 1,000 pF to 1,000,000 pF (4V)
	10	Size 0508	Chip (0805)	1,000 pF to 100,000 pF (50V) 1,000 pF to 100,000 pF (25V) 1,000 pF to 220,000 pF (16V) 1,000 pF to 1,000,000 pF (10V) 1,000 pF to 1,000,000 pF (6.3V) 1,000 pF to 2,200,000 pF (4V)
	11	Size 0508	Interdigitated chip (0508)	100,000 pF to 1,000,000 pF (4V)
	12	Size 0508	Interdigitated chip (0508)	330,000 pF to 2,200,000 pF (4V)
	13	Size 0508	Interdigitated, low profile, chip (0508)	100,000 pF to 330,000 pF (6.3V) 100,000 pF to 1,000,000 pF (4V)
MIL-PRF-32700 (Electrolytic, conductive polymer tantalum, surface mount, non-established reliability and established reliability)	1	Single anode	Molded single anode	330 µF (2.5V) 100 µF to 330 µF (3V) 68 µF to 470 µF (4V) 33 µF to 330 µF (6.3V) 22 µF to 150 µF (10V) 47 µF to 100 µF (16V) 22 µF to 47 µF (20V) 6.8 µF to 33 µF (25V) 22 µF to 33 µF (30V) 10 µF to 15 µF (35V) 10 µF µF (50V) 4.7 µF (63V)
	2	Multiple anode	Molded multiple anode	470 µF to 1,500 µF (2.5V) 470 µF to 1,500 µF (3V) 330 µF to 1,000 µF (4V) 220 µF to 680 µF (6.3V) 150 µF to 470 µF (10V) 150 µF to 330 µF (16V) 100 µF (20V) 68 µF to 100 µF (25V) 47 µF to 100 µF (30V) 33 µF to 47 µF (35V) 22 µF to 33 µF (50V) 10 µF to 15 µF (63V)
MIL-PRF-39001 (MICA, Established Reliability and Nonestablished Reliability)	5	CMR03	Radial lead, dipped	1.0 pF to 120 pF (300V) 15 pF to 200 pF (100V) 22 pF to 400 pF (50V)
	5	CMR04	Radial lead, dipped	1.0 pF to 240 pF (500V) 270 pF to 300 pF (300V) 330 pF to 390 pF (100V)
	5	CMR05	Radial lead, dipped	1.0 pF to 390 pF (500V)
	5	CMR06	Radial lead, dipped	430 pF to 4,700 pF (500V)
	5	CMR07	Radial lead, dipped	5,100 pF to 20,000 pF (500V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-39001 (Continued)	5	CMR08	Radial lead, dipped	22,000 pF to 51,000 pF (500V) 56,000 pF to 68,000 pF (300V) 75,000 pF to 91,000 pF (100V)
MIL-PRF-39003 (Electrolytic, solid electrolyte, tantalum, established reliability)	1	CSR13	Tubular, axial lead, hermetically sealed, polarized	0.0047 µF to 6.8 µF (100V) 0.1 µF to 15 µF (75V) 0.0047 µF to 22 µF (50V) 5.6 µF to 47 µF (35V) 1.2 µF to 100 µF (20V) 2.7 µF to 150 µF (15V) 3.9 µF to 220 µF (10V) 5.6 µF to 330 µF (6V)
	2	CSR09	Tubular, axial lead, hermetically sealed, polarized	0.047 µF to 1.2 µF (75V) 0.22 µF to 1.8 µF (50V) 0.33 µF to 2.7 µF (35V) 0.56 µF to 6.8 µF (20V) 1 µF to 8.2 µF (15V) 1.8 µF to 15 µF (10V) 2.7 µF to 18 µF (6V)
	3	CSR23	Tubular, axial lead, hermetically sealed, polarized	1.2 µF to 39 µF (50V) 1.8 µF to 68 µF (35V) 2.7 µF to 180 µF (20V) 4.7 µF to 330 µF (15V) 6.8 µF to 560 µF (10V) 10 µF to 1,000 µF (6V)
	4	CSR91	Tubular, axial lead, hermetically sealed, nonpolarized	0.0023 µF to 1.3 µF (100V) 0.34 µF to 7.5 µF (75V) 0.0023 µF to 11.0 µF (50V) 2.8 µF to 23.0 µF (35V) 0.6 µF to 50.0 µF (20V) 1.3 µF to 75.0 µF (15V) 1.9 µF to 110.0 µF (10V) 2.8 µF to 160.0 µF (6V)
	6	CSR33	Tubular, axial lead, hermetically sealed, polarized	1.2 µF to 39.0 µF (50V) 1.8 µF to 68.0 µF (35V) 2.7 µF to 180.0 µF (20V) 4.7 µF to 330.0 µF (15V) 6.8 µF to 560.0 µF (10V) 10.0 µF to 1000.0 µF (6V)
	9	CSR21	Tubular, axial lead, hermetically sealed, polarized, high frequency	5.6 µF to 22 µF (50V) 22 µF to 47 µF (35V) 27 µF to 100 µF (20V) 56 µF to 150 µF (15V) 82 µF to 220 µF (10V) 150 µF to 330 µF (6V)
	10	CSS13	Tubular, axial lead, hermetically sealed, polarized, high reliability	0.15 µF to 15 µF (75V) 0.12 µF to 22 µF (50V) 5.6 µF to 47 µF (35V) 1.2 µF to 100 µF (20V) 2.7 µF to 150 µF (15V) 3.9 µF to 220 µF (10V) 5.6 µF to 330 µF (6V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-39003 (Continued)	10	CSS33	Tubular, axial lead, hermetically sealed, polarized, high reliability	1.2 µF to 39 µF (50V) 1.8 µF to 68 µF (35V) 2.7 µF to 180 µF (20V) 4.7 µF to 330 µF (15V) 6.8 µF to 560 µF (10V) 10 µF to 1,000 µF (6V)
	11	CSS21	Tubular, axial lead, hermetically sealed, polarized, high reliability	5.6 µF to 22 µF (50V) 22 µF to 47 µF (35V) 27 µF to 100 µF (20V) 56 µF to 150 µF (15V) 82 µF to 220 µF (10V) 150 µF to 330 µF (6V)
MIL-PRF-39006 (Electrolytic, nonsolid electrolyte, tantalum, established reliability)	22	CLR79	Tubular, axial lead, hermetically sealed, polarized	1.7 µF to 56 µF (125V) 2.5 µF to 86 µF (100V) 3.5 µF to 110 µF (75V) 4 µF to 140 µF (60V) 5 µF to 160 µF (50V) 8 µF to 300 µF (30V) 10 µF to 350 µF (25V) 15 µF to 540 µF (15V) 20 µF to 750 µF (10V) 25 µF to 850 µF (8V) 30 µF to 1,200 µF (6V)
	25	CLR81	Tubular, axial lead, hermetically sealed, polarized, extended capacitance range	6.8 µF to 82 µF (125V) 10 µF to 120 µF (100V) 22 µF to 220 µF (75V) 27 µF to 270 µF (60V) 33 µF to 330 µF (50V) 56 µF to 560 µF (30V) 68 µF to 680 µF (25V) 100 µF to 1,000 µF (15V) 150 µF to 1,500 µF (10V) 180 µF to 1,800 µF (8V) 220 µF to 2,200 µF (6V)
	30	CLR90	Tubular, axial lead, hermetically sealed, polarized, low ESR	1.7 µF to 56 µF (125V) 2.5 µF to 86 µF (100V) 3.5 µF to 110 µF (75V) 4 µF to 140 µF (60V) 5 µF to 160 µF (50V) 8 µF to 300 µF (30V) 10 µF to 350 µF (25V) 15 µF to 540 µF (15V) 20 µF to 750 µF (10V) 25 µF to 850 µF (8V) 30 µF to 1,200 µF (6V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-39006 (Continued)	31	CLR91	Tubular, axial lead, hermetically sealed, polarized, extended capacitance range, low ESR	6.8 µF to 82 µF (125V) 10 µF to 120 µF (100V) 22 µF to 220 µF (75V) 27 µF to 270 µF (60V) 33 µF to 330 µF (50V) 56 µF to 560 µF (30V) 68 µF to 680 µF (25V) 100 µF to 1,000 µF (15V) 150 µF to 1,500 µF (10V) 180 µF to 1,800 µF (8V) 220 µF to 2,200 µF (6V)
	33	CLR93	Tubular, axial lead, hermetically sealed, polarized, extended capacitance range	10 µF to 150 µF (125V) 15 µF to 220 µF (100V) 33 µF to 470 µF (75V) 47 µF to 560 µF (60V) 68 µF to 680 µF (50V) 100 µF to 1,500 µF (30V) 120 µF to 1,800 µF (25V)
MIL-PRF-39014 (Ceramic, general purpose, established and nonestablished reliability)	1	CKR05	Molded, radial lead (.190 X .190)	10 pF to 1,000 pF (200V) 1,200 pF to 10,000 pF (100V) 12,000 pF to 100,000 pF (50V)
	2	CKR06	Molded, radial lead (.290 X .290)	1,200 pF to 10,000 pF (200V) 12,000 pF to 100,000 pF (100V) 120,000 pF to 1,000,000 pF (50V)
	5	CKR11	Tubular, axial lead (.160 X .090)	10 pF to 4,700 pF (100V) 5,600 pF to 10,000 pF (50V)
	5	CKR12	Tubular, axial lead (.250 X .090)	5,600 pF to 10,000 pF (100V) 12,000 pF to 47,000 pF (50V)
	5	CKR14	Tubular, axial lead (.390 X .140)	12,000 pF to 100,000 pF (100V) 56,000 pF to 270,000 pF (50V)
	5	CKR15	Tubular, axial lead (.500 X .250)	56,000 pF to 330,000 pF (100V) 470,000 pF to 1,000,000 pF (50V)
	5	CKR16	Tubular, axial lead (.690 X .350)	470,000 µF to 1,000,000 pF (100V) 2,200,000 pF to 3,300,000 pF (50V)
	20	CKR08	Molded, radial lead (.290 X .290)	1,200,000 pF to 2,000,000 pF (50V)
	22	CKR22	Molded, 2 pin DIP (.260 X .175)	1.0 pF to 820 pF (200V) 560 pF to 10,000 pF (100V) 2,700 pF to 100,000 pF (50V)
	22	CKR23	Molded, 2 pin DIP (.260 X .195)	560 pF to 10,000 pF (200V) 2,700 pF to 100,000 pF (100V) 4,700 pF to 220,000 pF (50V)
	22	CKR24	Molded, 2 pin DIP (.260 X .320)	120,000 pF to 150,000 pF (100V) 180,000 pF to 1,000,000 pF (50V)
	23	CKR04	Molded, radial lead (.190 X .190)	10 pF to 1,000 pF (200V) 1,200 pF to 10,000 pF (100V) 12,000 pF to 100,000 pF (50V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-39018 (Electrolytic, aluminum oxide, established and nonestablished reliability	1	CU13/CUR13	Tubular, axial lead, polarized	1.0 µF to 12 µF (350V) 1.5 µF to 18 µF (300V) 2.2 µF to 27 µF (250V) 3.3 µF to 39 µF (200V) 4.7 µF to 56 µF (150V) 8.2 µF to 68 µF (100V) 12 µF to 120 µF (75V) 22 µF to 180 µF (50V) 33 µF to 390 µF (30V) 68 µF to 680 µF (15V) 100 µF to 820 µF (10V) 220 µF to 1000 µF (7V)
	2	CU15	Tubular, axial lead, nonpolarized	.68 µF to 10 µF (250V) 1.0 µF to 15 µF (200V) 1.8 µF to 22 µF (150V) 2.2 µF to 33 µF (100V) 3.9 µF to 56 µF (75V) 6.8 µF to 82 µF (50V) 15 µF to 180 µF (30V) 33 µF to 390 µF (15V) 39 µF to 470 µF (10V) 47 µF to 680 µF (7V)
	3	CU17/CUR17	Tubular, axial lead, polarized	5.8 µF to 120 µF (350V) 8.2 µF to 150 µF (300V) 10 µF to 180 µF (250V) 15 µF to 270 µF (200V) 18 µF to 330 µF (150V) 22 µF to 470 µF (100V) 47 µF to 820 µF (75V) 100 µF to 1,800 µF (50V) 150 µF to 2,700 µF (30V) 390 µF to 6,800 µF (15V) 470 µF to 8,200 µF (10V) 680 µF to 12,000 µF (7V)
	4	CU71/CUR71	Tubular, large can, screw-insert terminals, polarized	68 µF to 2,200 µF (350V) 100 µF to 3,400 µF (300V) 100 µF to 3,300 µF (250V) 150 µF to 4,100 µF (200V) 220 µF to 4,700 µF (150V) 250 µF to 6,800 µF (100V) 530 µF to 14,000 µF (75V) 1,000 µF to 39,000 µF (50V) 1,800 µF to 48,000 µF (40V) 2,200 µF to 58,000 µF (30V) 2,200 µF to 68,000 µF (25V) 2,800 µF to 82,000 µF (20V) 3,300 µF to 110,000 µF (15V) 4,700 µF to 150,000 µF (10V) 6,800 µF to 220,000 µF (5V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-39018 (Continued)	6	CUR91	Tubular, large can, screw-insert terminals, polarized	210 µF to 4,700 µF (150V) 410 µF to 9,500 µF (100V) 820 µF to 19,000 µF (75V) 1,400 µF to 39,000 µF (50V) 1,800 µF to 48,000 µF (40V) 2,200 µF to 58,000 µF (30V) 2,500 µF to 67,000 µF (25V) 2,800 µF to 82,000 µF (20V) 4,000 µF to 110,000 µF (15V) 5,000 µF to 140,000 µF (10V) 8,000 µF to 220,000 µF (5V)
	7	CUR19	Tubular, four terminal, axial lead, polarized	50 µF to 320 µF (200V) 70 µF to 470 µF (150V) 130 µF to 850 µF (100V) 200 µF to 1,300 µF (75V) 300 µF to 2,000 µF (50V) 550 µF to 3,600 µF (40V) 700 µF to 4,500 µF (30V) 900 µF to 5,700 µF (25V) 1,200 µF to 8,600 µF (20V) 1,400 µF to 10,000 µF (16V) 1,700 µF to 12,000 µF (10V) 2,000 µF to 14,000 µF (7.5V) 2,400 µF to 16,000 µF (5V)
MIL-PRF-39022 (Metallized, paper-plastic or plastic film, direct and alternating current, hermetically sealed, established reliability	1	CHR09	Tubular, axial lead, DC	.01 µF to 2.2 µF (600V) .047 µF to 3.3 µF (400V) .1 µF to 12 µF (200V) .018 µF to 12 µF (50V)
	2	CHR19	Tubular, axial lead, DC	.047 µF to 3.3 µF (400V) .1 µF to 8.2 µF (200V)
	8	CHR12	Tubular, axial lead, DC	.01 µF to 2.2 µF (600V) .047 µF to 3.3 µF (400V) .1 µF to 12 µF (200V) .15 µF to 12 µF (50V)
	12	CHR26	Tubular, axial lead, AC	.01 µF to .47 µF (300 - 400Vac) 0.56 µF to 2.2 µF (200 - 295Vac) 0.1 µF to 8 µF (100 - 165Vac) 9 µF to 10µF (80 - 90Vac)
	13	CHR27	Tubular, axial lead, DC	.01 µF to 1 µF (600V) .01 µF to 2.7 µF (400V) .01 µF to 3.9 µF (300V) .01 µF to 10 µF (200V) .01 µF to 22 µF (100V) .47 µF to 22 µF (50V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-49137 (Electrolytic, solid electrolyte, tantalum, nonhermetically sealed, nonestablished reliability)	1	CX02	Dipped, tubular, axial lead, polarized, nonhermetically sealed	.1 µF to 5.6 µF (50V) .45 µF to 10 µF (35V) 1.5 µF to 12 µF (25V) 10 µF to 18 µF (20V) 15 µF to 33 µF (15V) 2.7 µF to 56 µF (10V) 3.9 V to 68 µF (6V)
	2	CX02/CX12	Dipped, tubular, axial lead, polarized, nonhermetically sealed	.1 µF to 22 µF (50V) 6.8 µF to 47 µF (35V) 1.5 µF to 68 µF (25V) 2.2 µF to 100 µF (20V) 3.3 µF to 150 µF (15V) 4.7 µF to 220 µF (10V) 6.8 µF to 330 µF (6V)
	5	CX05	Molded, tubular, axial lead, polarized, nonhermetically sealed	.1 µF to 4.7 µF (50V) .33 µF to 10 µF (35V) 1 µF to 10 µF (25V) 1.5 µF to 15 µF (20V) 2.2 µF to 33 µF (15V) 3.3 µF to 47 µF (10V) 4.7 µF to 47 µF (6V)
	6	CX06/CX16	Molded, rectangular, radial lead, polarized, nonhermetically sealed	.1 µF to 22 µF (35V) .68 µF to 33 µF (25V) .1 µF to 47 µF (20V) .1 µF to 68 µF (15V) .15 µF to 100 µF (10V) .22 µF to 150 µF (6V) .33 µF to 68 µF (4V) 1.5 µF to 220 µF (3V) .47 µF to 10 µF (2V)
MIL-PRF-49467 (Fixed, ceramic, multilayer, high voltage, general purpose)	1	HV01	Conformally coated, molded, or encapsulated	10 pF to .47 µF (1000V)
	2	HV02	Conformally coated, molded, or encapsulated	10 pF to .10 µF (2000V)
	3	HV03	Conformally coated, molded, or encapsulated	10 pF to .082 µF (3000V)
	4	HV04	Conformally coated, molded, or encapsulated	10 pF to .047 µF (4000V)
	5	HV05	Conformally coated, molded, or encapsulated	10 pF to .027 µF (5000V)
	7	HV07	Conformally coated, molded, or encapsulated	10 pF to .47 µF (600V)
MIL-PRF-49470 (Ceramic, switch mode power supply, general purpose and temperature stable)	1	PS01	Horizontally stacked, unencapsulated	.01 µF to 39 µF (500V) .022 µF to 120 µF (200V) .047 µF to 180 µF (100V) .056 µF to 270 µF (50V)
	2	PS02	Horizontally stacked, encapsulated	.01 µF to 39 µF (500V) .022 µF to 120 µF (200V) .047 µF to 180 µF (100V) .056 µF to 270 µF (50V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-55365 (Electrolytic, tantalum, chip, established reliability)	4	CWR06	Conformally coated tantalum chips	0.1 µF to 4.7 µF (50V) 0.22 µF to 6.8 µF (35V) 0.33 µF to 15 µF (25V) 0.47 µF to 22 µF (20V) 0.68 µF to 33 µF (15V) 1.0 µF to 47 µF (10V) 1.5 µF to 68 µF (6V) 2.2 µF to 100 µF (4V)
	4	CWR09	Molded tantalum chips	0.1 µF to 4.7 µF (50V) 0.22 µF to 6.8 µF (35V) 0.33 µF to 15 µF (25V) 0.47 µF to 22 µF (20V) 0.68 µF to 33 µF (15V) 1.0 µF to 47 µF (10V) 1.5 µF to 68 µF (6V) 2.2 µF to 100 µF (4V)
	8	CWR11	Molded tantalum chips (metric)	0.1 µF to 4.7 µF (50V) 0.1 µF to 6.8 µF (35V) 0.33 µF to 15 µF (25V) 0.47 µF to 22 µF (20V) 0.68 µF to 33 µF (15V) 1.0 µF to 47 µF (10V) 1.5 µF to 68 µF (6V) 2.2 µF to 100 µF (4V)
	11	CWR19	Molded tantalum chips	0.33 µF to 15 µF (35V) 0.47 µF to 33 µF (25V) 0.68 µF to 47 µF (20V) 1 µF to 100 µF (15V) 2.2 µF to 220 µF (10V) 3.3 µF to 330 µF (6V) 3.3 µF to 330 µF (4V)
	11	CWR29	Molded tantalum chips	0.1 µF to 4.7 µF (50V) 0.22 µF to 15 µF (35V) 0.33 µF to 33 µF (25V) 0.47 µF to 47 µF (20V) 0.68 µF to 100 µF (15V) 1 µF to 220 µF (10V) 1.5 µF to 330 µF (6V) 2.2 µF to 330 µF (4V)
	12	CWR15	Molded tantalum chips	0.1 µF to 1.5 µF (50V) 0.22 µF to 2.2 µF (35V) 0.33 µF to 4.7 µF (25V) 0.47 µF to 10 µF (20V) 1 µF to 15 µF (15V) 0.47 µF to 33 µF (10V) 3.3 µF to 68 µF (6V) 6.8 µF to 100 µF (4V)
	13	CWR16	Conformally coated tantalum chips	.33 µF to 15 µF (35V) .47 µF to 33 µF (25V) .68 µF to 47 µF (20V) 1 µF to 100 µF (15V) 2.2 to 220 µF (10V) 3.3 to 330 µF (6V) 3.3 µF to 330 (4V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
	1	CFR02	Tubular, axial lead, DC, non-hermetically sealed	.001 µF to .1 µF (600V) .0018 µF to 1 µF (400V) .001 µF to 1 µF (200V) .001 µF to 1 µF (100V)
	4	CFR05/CFR06	Tubular, axial lead, DC, non-hermetically sealed	.01 µF to 5 µF (400V) .01 µF to 10 µF (200V) .001 µF to 20 µF (100V) .001 µF to 50 µF (50V)
MIL-PRF-55514 (Plastic or metallized plastic, DC or DC-AC, established reliability)	9	CFR13/CFR14	Tubular, axial lead, DC, non-hermetically sealed	1 µF to 10 µF (400V) 1 µF to 20 µF (200V) 1 µF to 30 µF (100V)
	10	CFR15	Tubular, axial lead, DC, non-hermetically sealed	.0056 µF to .56 µF (800V) .001 µF to 1 µF (600V) .0068 µF to 1 µF (400V) .012 µF to 1 µF (200V)
	13	CFR30	Tubular, axial lead, DC or AC, non-hermetically sealed	.01 µF to 3.9 µF (200V) .015 µF to 5 µF (150V) .027 µF to 10 µF (100V) .033 µF to 15 µF (75V) .056 µF to 18 µF (50V)
MIL-PRF-55681 (Ceramic, chip, multiple layer, established and Nonestablished reliability)	1	CDR01	Chip (0805)	10 pF to 3,300 pF (100V) 3,900 pF to 4,700 pF (50V)
	1	CDR02	Chip (1805)	220 pF to 10,000 pF (100V) 12,000 pF to 22,000 pF (50V)
	1	CDR03	Chip (1808)	330 pF to 33,000 pF (100V) 39,000 pF to 68,000 pF (50V)
	1	CDR04	Chip (1812)	1,200 pF to 56,000 pF (100V) 82,000 pF to 180,000 pF (50V)
	2	CDR05	Chip (1825)	3,900 pF to 150,000 pF (100V) 220,000 pF to 330,000 pF (50V)
	3	CDR06	Chip (2225)	6,800 pF to 10,000 pF (100V) 390,000 pF to 470,000 pF (50V)
	4	CDR11/CDR12	Chip, high frequency (0505)	0.1 pF to 100 pF (150V) 0.1 pF to 1,000 pF (50V)
	4	CDR13/CDR14	Chip, high frequency (1111)	0.1 pF to 100 pF (500V) 110 pF to 200 pF (300V) 0.1 pF to 470 pF (200V) 510 pF to 620 pF (100V) 680 pF to 5,100 pF (50V)
	5	CDR21/CDR22/ CDR23/CDR24/ CDR25	Chip, high frequency (1411)	0.1 pF to 100 pF (500V) 110 pF to 200 pF (300V) 220 pF to 470 pF (200V) 510 pF to 620 pF (100V) 680 pF to 5,100 pF (50V)
	7	CDR31	Metric chip (2.0 mm X 1.25 mm) (0805)	1 pF to 4,700 pF (100V) 510 pF to 18,000 pF (50V)
	8	CDR32	Metric chip (3.2 mm X 1.6 mm) (1206)	1 pF to 15,000 pF (100V) 1,110 pF to 39,000 pF (50V)
	9	CDR33	Metric chip (3.2 mm X 2.5 mm) (1210)	1,000 pF to 27,000 pF (100V) 2,400 pF to 100,000 pF (50V)
	10	CDR34	Metric chip (4.5 mm X 3.2 mm) (1812)	2,200 pF to 56,000 pF (100V) 5,100 pF to 180,000 pF (50V)
	11	CDR35	Metric chip (4.5 mm X 6.4 mm) (1825)	4,700 pF to 150,000 pF (100V) 11,000 pF to ,470,000 pF (50V)

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TABLE A-II. Capacitor selection guidance table - continued.

Specification	Spec. sheet	Styles	Description (dimension - inches)	Available capacitance range (Voltage)
MIL-PRF-55681 (Continued)	12	CDR36	Chip (0603)	100 pF to 1,000 pF (200V) 0.5 pF to 8,200 pF (100V) 0.5 pF to 27,000 pF (50V) 0.5 pF to 56,000 pF (25V) 0.5 pF to 100,000 pF (16V) 0.5 pF to 100,000 pF (10V) 0.5 pF to 100,000 pF (6.3V) 0.5 pF to 100,000 pF (4V)
	13	CDR37	Chip (0402)	0.5 pF to 1,000 pF (100V) 0.5 pF to 3,900 pF (50V) 0.5 pF to 4,700 pF (25V) 0.5 pF to 10,000 pF (16V) 0.5 pF to 10,000 pF (10V) 0.5 pF to 10,000 pF (6.3V) 0.5 pF to 10,000 pF (4V)
MIL-PRF-83421 (Metallized, plastic film, DC, AC, or DC and AC, hermetically sealed, established reliability)	1	CRH01	Axial lead, hermetically sealed	0.001 µF to 22 µF (30V)
	1	CRH02	Axial lead, hermetically sealed	0.001 µF to 10 µF (50V)
	1	CRH03	Axial lead, hermetically sealed	0.001 µF to 10 µF (100V)
	1	CRH04	Axial lead, hermetically sealed	0.001 µF to 3.9 µF (200V)
	1	CRH05	Axial lead, hermetically sealed	0.001 µF to 2.0 µF (400V)
	2	CRH11	Axial lead, hermetically sealed	0.47 µF to 25 µF (100V)
	2	CRH12	Axial lead, hermetically sealed	0.18 µF to 5 µF (200V)
	2	CRH13	Axial lead, hermetically sealed	0.001 µF to 2.0 µF (400V)
	6	CRH31	Axial lead, hermetically sealed	0.001 µF to 22 µF (30V)
	6	CRH32	Axial lead, hermetically sealed	0.001 µF to 10 µF (50V)
	6	CRH33	Axial lead, hermetically sealed	0.001 µF to 10 µF (100V)
	6	CRH34	Axial lead, hermetically sealed	0.001 µF to 3.9 µF (200V)
	6	CRH35	Axial lead, hermetically sealed	0.001 µF to 2.0 µF (400V)

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TABLE A-III. Principal capacitor applications.

MIL-PRF-	ER	Type	Blocking	Buffering	By-passing	Coupling	Filtering	Tuning	Temp. compen-sating	Trimming	Motor starting	Timing	Noise suppression
20	X	Ceramic			X	X	X	X	X				
81		Ceramic Trimmer		X		X		X		X			
123		Ceramic			X	X	X						
11693	X	Plastic					X						X
14409		Piston Trimmer						X		X			
19978	X	Plastic	X	X	X	X	X						
32535		Ceramic			X	X	X						
32700	X	Polymer Tantalum			X	X	X						
39001	X	Mica	X	X	X	X	X	X				X	
39003	X	Solid Tantalum	X		X	X	X						X
39006	X	Wet Tantalum	X		X	X	X				X		X
39014	X	Ceramic			X	X	X						
39018	X	Aluminum	X		X	X	X						
39022	X	Metallized Plastic	X		X		X						
49137	X	Solid Tantalum	X		X	X	X						
49467		Ceramic, High Voltage	X			X	X						
49470		Ceramic, SMPS	X	X	X	X	X	X		X			X
55365	X	Solid Tantalum Chip			X	X	X						
55514	X	Plastic	X		X								
55681	X	Ceramic Chip			X	X	X	X					
83421	X	Metallized Plastic	X	X	X	X	X	X			X	X	

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TABLE A-IV. Detailed specification number by style number.

Performance specification	Style	Performance specification	Style
MIL-PRF-20/27	CCR75	MIL-PRF-14409/16	PC23
MIL-PRF-20/29	CCR77	MIL-PRF-14409/16	PC24
MIL-PRF-20/30	CCR78	MIL-PRF-14409/17	PC17
MIL-PRF-20/31	CCR79	MIL-PRF-14409/18	PC18
MIL-PRF-20/35	CCR05	MIL-PRF-14409/18	PC46
MIL-PRF-20/35	CCR09	MIL-PRF-14409/19	PC19
MIL-PRF-20/36	CCR06	MIL-PRF-14409/19	PC47
MIL-PRF-20/37	CCR07	MIL-PRF-19978/1	CQ08
MIL-PRF-20/38	CCR08	MIL-PRF-19978/1	CQ09
MIL-PRF-81/1	CV11	MIL-PRF-19978/1	CQ12
MIL-PRF-81/4	CV31	MIL-PRF-19978/1	CQ13
MIL-PRF-81/4	CV32	MIL-PRF-19978/2	CQ20
MIL-PRF-81/6	CV35	MIL-PRF-19978/3	CQ72
MIL-PRF-81/6	CV36	MIL-PRF-19978/9	CQR09
MIL-PRF-123/1	CKS05	MIL-PRF-19978/10	CQR12
MIL-PRF-123/2	CKS06	MIL-PRF-19978/11	CQR13
MIL-PRF-123/3	CKS07	MIL-PRF-19978/13	CQR29
MIL-PRF-123/4	CKS11	MIL-PRF-19978/14	CQR32
MIL-PRF-123/5	CKS12	MIL-PRF-19978/15	CQR33
MIL-PRF-123/6	CKS14	MIL-PRF-19978/22	CQR44
MIL-PRF-123/7	CKS15	MIL-PRF-32535/1	Size 0201
MIL-PRF-123/8	CKS16	MIL-PRF-32535/2	Size 0402
MIL-PRF-123/10	CKS51	MIL-PRF-32535/3	Size 0603
MIL-PRF-123/11	CKS52	MIL-PRF-32535/4	Size 0805
MIL-PRF-123/12	CKS53	MIL-PRF-32535/5	Size 1206
MIL-PRF-123/13	CKS54	MIL-PRF-32535/6	Size 1210
MIL-PRF-123/16	CKS22	MIL-PRF-32535/7	Size 1812
MIL-PRF-123/17	CKS23	MIL-PRF-32535/8	Size 2220
MIL-PRF-123/18	CKS24	MIL-PRF-32535/9	Size 0306
MIL-PRF-123/21	CKS55	MIL-PRF-32535/10	Size 0508
MIL-PRF-123/22	CKS56	MIL-PRF-32535/11	Size 0508
MIL-PRF-123/23	CKS57	MIL-PRF-32535/12	Size 0508
MIL-PRF-11693/7	CZ23, CZR23	MIL-PRF-32535/13	Size 0508
MIL-PRF-11693/7	CZ24, CZR24	MIL-PRF-32700/1	Single anode
MIL-PRF-14409/4	PC38	MIL-PRF-32700/2	Multiple anode
MIL-PRF-14409/4	PC48	MIL-PRF-39001/5	CMR03
MIL-PRF-14409/5	PC39	MIL-PRF-39001/5	CMR04
MIL-PRF-14409/6	PC40	MIL-PRF-39001/5	CMR05
MIL-PRF-14409/6	PC50	MIL-PRF-39001/5	CMR06
MIL-PRF-14409/7	PC41	MIL-PRF-39001/5	CMR07
MIL-PRF-14409/7	PC51	MIL-PRF-39001/5	CMR08
MIL-PRF-14409/8	PC42	MIL-PRF-39003/1	CSR13
MIL-PRF-14409/8	PC52	MIL-PRF-39003/2	CSR09
MIL-PRF-14409/9	PC43	MIL-PRF-39003/3	CSR23
MIL-PRF-14409/12	PC25	MIL-PRF-39003/4	CSR91
MIL-PRF-14409/12	PC31	MIL-PRF-39003/6	CSR33
MIL-PRF-14409/13	PC26	MIL-PRF-39003/9	CSR21
MIL-PRF-14409/14	PC27	MIL-PRF-39003/10	CSS13
MIL-PRF-14409/14	PC28	MIL-PRF-39003/10	CSS33
MIL-PRF-14409/14	PC29	MIL-PRF-39003/11	CSS21
MIL-PRF-14409/15	PC30	MIL-PRF-39006/22	CLR79
MIL-PRF-14409/15	PC32	MIL-PRF-39006/25	CLR81
MIL-PRF-14409/15	PC34	MIL-PRF-39006/30	CLR90
MIL-PRF-14409/16	PC21	MIL-PRF-39006/31	CLR91
MIL-PRF-14409/16	PC22	MIL-PRF-39006/33	CLR93

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TABLE A-IV. Detailed specification number by style number - continued.

Performance specification	Style	Performance specification	Style
MIL-PRF-39014/1	CKR05	MIL-PRF-55514/1	CFR02
MIL-PRF-39014/2	CKR06	MIL-PRF-55514/4	CFR05
MIL-PRF-39014/5	CKR11	MIL-PRF-55514/4	CFR06
MIL-PRF-39014/5	CKR12	MIL-PRF-55514/9	CFR13
MIL-PRF-39014/5	CKR14	MIL-PRF-55514/9	CFR14
MIL-PRF-39014/5	CKR15	MIL-PRF-55514/10	CFR15
MIL-PRF-39014/5	CKR16	MIL-PRF-55514/13	CFR30
MIL-PRF-39014/20	CKR08	MIL-PRF-55681/1	CDR01
MIL-PRF-39014/22	CKR22	MIL-PRF-55681/1	CDR02
MIL-PRF-39014/22	CKR23	MIL-PRF-55681/1	CDR03
MIL-PRF-39014/22	CKR24	MIL-PRF-55681/1	CDR04
MIL-PRF-39014/23	CKR04	MIL-PRF-55681/2	CDR05
MIL-PRF-39018/1	CU13, CUR13	MIL-PRF-55681/3	CDR06
MIL-PRF-39018/2	CU15	MIL-PRF-55681/4	CDR11
MIL-PRF-39018/3	CU17, CUR17	MIL-PRF-55681/4	CDR12
MIL-PRF-39018/4	CU71, CUR71	MIL-PRF-55681/4	CDR13
MIL-PRF-39018/6	CUR91	MIL-PRF-55681/4	CDR14
MIL-PRF-39018/7	CUR19	MIL-PRF-55681/5	CDR21
MIL-PRF-39022/1	CHR09	MIL-PRF-55681/5	CDR22
MIL-PRF-39022/2	CHR19	MIL-PRF-55681/5	CDR23
MIL-PRF-39022/8	CHR12	MIL-PRF-55681/5	CDR24
MIL-PRF-39022/12	CHR26	MIL-PRF-55681/5	CDR25
MIL-PRF-39022/13	CHR27	MIL-PRF-55681/7	CDR31
MIL-PRF-49137/1	CX01	MIL-PRF-55681/8	CDR32
MIL-PRF-49137/2	CX02	MIL-PRF-55681/9	CDR33
MIL-PRF-49137/2	CX12	MIL-PRF-55681/10	CDR34
MIL-PRF-49137/5	CX05	MIL-PRF-55681/11	CDR35
MIL-PRF-49137/6	CX06, CX16	MIL-PRF-55681/12	CDR36
MIL-PRF-49467/1	HV01	MIL-PRF-55681/13	CDR37
MIL-PRF-49467/2	HV02	MIL-PRF-83421/1	CRH01
MIL-PRF-49467/3	HV03	MIL-PRF-83421/1	CRH02
MIL-PRF-49467/4	HV04	MIL-PRF-83421/1	CRH03
MIL-PRF-49467/5	HV05	MIL-PRF-83421/1	CRH04
MIL-PRF-49467/7	HV07	MIL-PRF-83421/1	CRH05
MIL-PRF-49470/1	PS01	MIL-PRF-83421/2	CRH11
MIL-PRF-49470/2	PS02	MIL-PRF-83421/2	CRH12
MIL-PRF-55365/4	CWR06	MIL-PRF-83421/2	CRH13
MIL-PRF-55365/4	CWR09	MIL-PRF-83421/6	CRH31
MIL-PRF-55365/8	CWR11	MIL-PRF-83421/6	CRH32
MIL-PRF-55365/11	CWR19, CWR29	MIL-PRF-83421/6	CRH33
MIL-PRF-55365/12	CWR15	MIL-PRF-83421/6	CRH34
MIL-PRF-55365/13	CWR16	MIL-PRF-83421/6	CRH35

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